

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: FOR INFORMATION ONLY
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____

CERTIFIED BY CQCM or Designee 

02600 - Concrete Manholes - specification section

SECTION 02600
CONCRETE MANHOLES

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. Manhole frames and covers.
 - 2. Manhole grade rings.
 - 3. Manholes cones and risers.
 - 4. Manhole bases.

1.02 REFERENCES

- A. American Association of State Highway and Transportation Officials (AASHTO).
 - 1. Standard Specifications for Highway Bridges.
- B. ASTM International (ASTM):
 - 1. A48 - Standard Specification for Gray Iron Castings.
 - 2. C33 - Standard Specification for Concrete Aggregates.
 - 3. C150 - Standard Specification for Portland Cement.
 - 4. C443 - Standard Specification for Joints for Concrete Pipe and Manholes Using Rubber Gaskets.
 - 5. C478 - Standard Specification for Circular Precast Reinforced Concrete Manhole Sections.
 - 6. C990 - Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint.
 - 7. C923 - Standard Guide for In-Plant Performance Evaluation of Automatic Pedestrian SNM Monitors.
 - 8. C1244 - Standard Test Method for Concrete Sewer Manholes by the Negative Air Pressure (Vacuum) Test Prior to Backfill.
- C. International Organization of Standardization (ISO):
 - 1. 9001 - Quality Management Standard.

1.03 SUBMITTALS

- A. Submit as specified in Section 01330 - Submittal Procedures.

-  B. Shop drawings:
 -  1. Manufacturer's catalog data and details of following items for approval:
 - a. Frame and covers.
 - b. Grade rings.
 - c. Manhole cones and risers.
 - d. Manhole bases, if precast.

Nan, Inc. will submit
this at a later date

- 2. Manhole construction details, jointing methods, connection details, materials, and dimensions.
- 3. Repair procedures and details.
- C. Calculations and criteria used in manhole design including material properties, loadings, load combinations, and dimensions assumed.
- D. Test methods and results including certification that the manhole riser exceeds the minimum requirements in accordance with ASTM C478.
- E. Sealed drawings and design calculations by a licensed Professional Engineer licensed in the State where the project is located.
- F. Certificates:
 - 1. ISO 9001 certificate by a third party confirming that ASTM test reports are valid and up to date at the time of the bid and during construction period.
 - 2. Manufacturer's Certificate of Source Testing.

1.04 PRODUCT REQUIREMENTS

- A. As specified in Section 01600 - Product Requirements.
- B. Provide suitable quantities of lifting equipment to handle the manholes/risers and castings.
 - 1. In no case shall any equipment be used that is not rated to handle the intended loading or conditions of use to which it will be subjected, or which will damage or gouge the manhole components.
 - 2. Dragging or dropping the manhole components shall not be allowed.
- C. Source testing.
 - 1. Perform pre-production and post-production tests by manufacturer staff with a minimum of 5 years of experience in quality control, inspection, and testing of manholes.
 - a. In lieu of this experience, witness of tests by up to 3 full-time Owner representatives.
 - 2. Examine each completed manhole section for dimensional requirements, strength, and workmanship.
 - 3. Complete required testing in accordance with ASTM C478.
 - 4. Provide the Manufacturer's Certificate of Source Testing.

1.05 DESIGN CRITERIA:

- A. Manholes shall not include steps.
- B. Manhole lids: Locking type.
- C. Manhole bases:
 - 1. Constructed as part of the pipe installation or using tee risers.

- D. Manhole riser:
 - 1. Made of the same pipe material selected for the project, providing a sealed connection between the pipe base and riser as indicated on the Drawings.
 - 2. Manufactured specifically for this project and no materials shall be furnished from stock unless approved by the Engineer.
- E. Manhole provider shall coordinate with the pipe manufacturer for dimensions and connections.
- F. Manhole systems:
 - 1. Provided by a single manufacturer.
- G. Frames and covers:
 - 1. Provided by a single manufacturer unless approved by the Engineer.

1.06 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Manhole grade rings, cones, and risers: One of the following or equal.
 - 1. Jensen Precast.
 - 2. Hawaii Precast.
- B. Manhole frames and covers: Manufacturers: One of the following or equal.
 - 1. Alhambra Foundry Company.
 - 2. Neenah Foundry Company.

2.02 BASIS OF DESIGN

- A. Design in accordance with local jurisdiction requirements, including but not limited to the following:
 - 1. Manhole frames and covers.
 - 2. Manhole diameter tolerance.
 - 3. Manhole length.
 - 4. Manhole cover bolting.
 - 5. Manhole backfill.
- B. Structural design calculations:
 - 1. Load rating of manhole:
 - a. Design to support an AASHTO Standard Specifications for Highway Bridges, H-20 vehicle loading.
 - 2. Resist buoyancy:
 - a. Design with sufficient bottom anchorage and side friction to resist buoyancy.
 - b. Depths will be as indicated on the Drawings.
 - c. Design for fully saturated soil conditions.

- C. Manholes, grade rings, risers, and bases:
 - 1. Manhole:
 - a. Nominal sizes as indicated on the Drawings.
 - 2. Grade rings:
 - a. At least 1 but not more than 2 grade rings shall be used.
 - b. Maximum total distance from top of cone section to final grade: 18 inches.
 - 3. Cone and riser sections:
 - a. As specified in this Section and as indicated on the Drawings.
 - b. Cone sections shall be eccentric.
 - 4. Manhole bases:
 - a. As specified in this Section and as indicated on the Drawings.
 - b. Provide corrosion protection system on unlined concrete.
- D. Threaded lifting inserts:
 - 1. Design inserts to be fully threaded:
 - a. Do not fully penetrate through entire manhole wall.
 - 2. Provide lifting device compatible with spreader bar and chains, hooks, and slings.
 - 3. Design with minimum safety factor of 4.0.
 - 4. Do not use reinforcing steel bars.

2.03 MATERIALS

- A. Cast iron manhole frames and covers.
 - 1. In accordance with ASTM A48.
 - 2. Covers: Bolted as specified.
 - 3. Concrete collars: As indicated on the Drawings.
- B. Manhole bases, risers, and grade rings:
 - 1. Cement: Type II or Type V portland cement in accordance with ASTM C150.
 - 2. Concrete aggregates: In accordance with ASTM C33, gradation as specified in approved mix design.
 - 3. Sections: Steel reinforced.
 - 4. Precast concrete sections: Manufactured by a process that will produce a dense, homogeneous concrete ring.
 - 5. Top and bottom of sections: Parallel.
- C. Joint sealant:
 - 1. Preformed, cold applied flexible joint sealant in accordance with ASTM C990 and ASTM C443.
 - 2. Manufacturers: One of the following or equal.
 - a. Henry Corp., Ram-Nek.
 - b. Kent Seal - Hamilton Kent Corp.

2.04 COMPONENTS

- A. Pipe stubs:
 - 1. Provide pipe stubs at manhole locations and in accordance with details indicated on the Drawings and as specified.
 - 2. Plugging stubs:
 - a. Plug stubs with vitrified clay stopper, brick plug, or other materials as indicated on the Drawings.
 - b. Unless otherwise indicated on the Drawings, comply with following:
 - 1) Stubs up to and including 21 inches: Vitrified clay or PVC stoppers.
 - 2) Stubs greater than 21 inches: Brick plugs.
- B. Resilient pipe connectors:
 - 1. Unless otherwise indicated on the Drawings or specified, provide a flexible compression type connector between manhole and pipes entering and leaving the manhole in accordance with ASTM C923.
 - 2. Resilient pipe connectors:
 - a. Manufacturers: The following or equal.
 - 1) A-LOK Premium.
 - 2) Provide 2 flexible joints within 4 feet of manhole.
- C. Threaded lifting inserts.
- D. Drop manholes:
 - 1. Construct drop manholes at locations and in accordance with details indicated on the Drawings.
 - 2. Provide inside diameter of drop inlet pipe the same as intercepted sewer unless otherwise indicated on the Drawings or specified in this Section.
 - 3. Furnish and set fittings as indicated on the Drawings.

2.05 IDENTIFICATION MARKINGS

- A. Identification marks on the exterior of bases, risers, grade rings, and include the following information:
 - 1. Date of manufacture of the item.
 - 2. Name or trademark of the manufacturer.
 - 3. Internal diameter in inches.
 - 4. Number of the manhole as indicated on the Drawings.

2.06 QUALITY CONTROL

- A. Manufacturer to provide permanent quality control department and laboratory facility capable of performing inspections and testing as specified by this Section.
- B. Material testing, inspection procedures, and manufacturing process are subject to inspection by the Owner or Owner's representative.
- C. Perform manufacturer's tests and inspections in accordance with the referenced standards and as specified in this Section including the following:
 - 1. Provide the Manufacturer's Certificate of Source Testing as specified in Section 01600 - Product Requirements.

2. Manufacturer shall make available services of representative throughout the project duration when deemed necessary by the Engineer.
 3. Calibrate within last 12 months for equipment such as scales, measuring devices and calibration tools used in the manufacturing of pipe.
 - a. Each device used in the manufacture of manholes is required to have a tag recording date of last calibration.
 - b. Devices are subject to inspection by Engineer.
- D. Furnish labor necessary to assist the Engineer in inspecting manholes upon delivery.
- E. Remove rejected manholes immediately.

PART 3 EXECUTION

3.01 MANHOLE INSTALLATION

- A. Excavate and backfill as specified in Section 02318 - Trenching and as indicated on the Drawings.
- B. Maintain identification markings on installed pieces throughout installation.
- C. Do not use sections with chips or cracks in the joint.
- D. Engineer may inspect manhole sections, prior to installation.
- E. Repair of manhole sections damaged during installation in accordance with manufacturer's repair procedures; with the concurrence of the Engineer.
- F. Bedding and backfill for tee-base pipe fitting: In accordance with Section 02318 - Trenching.
 1. Tee-base outlet: Vertical.
 2. Upper end of the vertical pipe stub creates a horizontal plane.
- G. Tee bases:
 1. Cast-in-place reinforced concrete as indicated on the Drawings.
 2. Vibrate to densify concrete and screed so first precast manhole riser section has a level, uniform bearing for full circumference.
 3. Use steel impression ring to create a groove into which the first manhole riser section tongue can be inserted.
 4. Install preformed plastic sealing gasket between groove and tongue and first manhole riser section.
- H. Install joint sealant material in accordance with manufacturer's instructions:
 1. Completed manhole: Rigid and watertight.
- I. Fill threaded lifting inserts with grout.

- J. Lay grade rings on joint sealant with sides plumb and tops level.
- K. Set frame and covers as specified and as indicated on the Drawings.
- L. For Tee base manholes, apply dampproofing as specified in Section 07110 - Dampproofing and as indicated on the Drawings.
 - 1. Apply material in accordance with manufacturer recommendations over the entire exterior surface of the completed manhole, including base section, riser sections, cone section, and grade rings prior to backfilling the manhole.
 - 2. Allow waterproofing material to dry sufficiently before backfilling.
- M. After installation coat all interior concrete surfaces of manholes, including channels and shelves, with an elastomeric polyurethane per coating system POL-C-1-BSC as specified in Section 09960 - High-Performance Coatings.

3.02 CLEANING

- A. After completing each manhole, remove debris, construction materials, and equipment from the site of the work, grade, and smooth over the surface and leave the entire right-of-way in a clean, neat, and serviceable condition.
- B. After completing each manhole, remove construction material debris from inside the manhole.

3.03 FUNCTIONAL TESTING

- A. Provide materials for grouting and patching recommended by the manufacturer or an approved equal.
- B. Vacuum testing in accordance with ASTM C1244.
 - 1. Install the vacuum test head on top of the manhole.
 - a. Install and brace sealing devices on influent and effluent pipes.
 - 2. Draw a vacuum of 10 inches of mercury with a vacuum pump, deactivate the pump, and measure the actual elapsed time for the vacuum to drop to 9 inches of mercury.

3. Compare test results with the minimum time requirements stated in the table below.
 - a. If the actual elapsed time is less than the time in the table, the manhole is defective, and it shall be repaired and retested until it is acceptable.

<u>Minimum Elapsed Time, Minutes: Seconds</u>				
Manhole Depth, ft.	Manhole Diameter, inches			
	36	48	60	72
8	0:14	0:20	0:26	0:33
10	0:18	0:25	0:33	0:41
12	0:21	0:30	0:39	0:49
14	0:25	0:35	0:46	0:57
16	0:28	0:40	0:52	1:07
18	0:32	0:45	0:59	1:13
20	0:35	0:50	1:05	1:21
22	0:38	0:55	1:12	1:29
24	0:42	0:59	1:18	1:37
26	0:46	1:04	1:25	1:45
28	0:49	1:09	1:31	1:53
30	0:53	1:14	1:38	2:01

END OF SECTION

Concrete Manholes - Submittal package

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-1**
Type: MH
Size: 48"
Rim: 38.17'
Invert: 30.7'
Rim to Invert: 7.47'

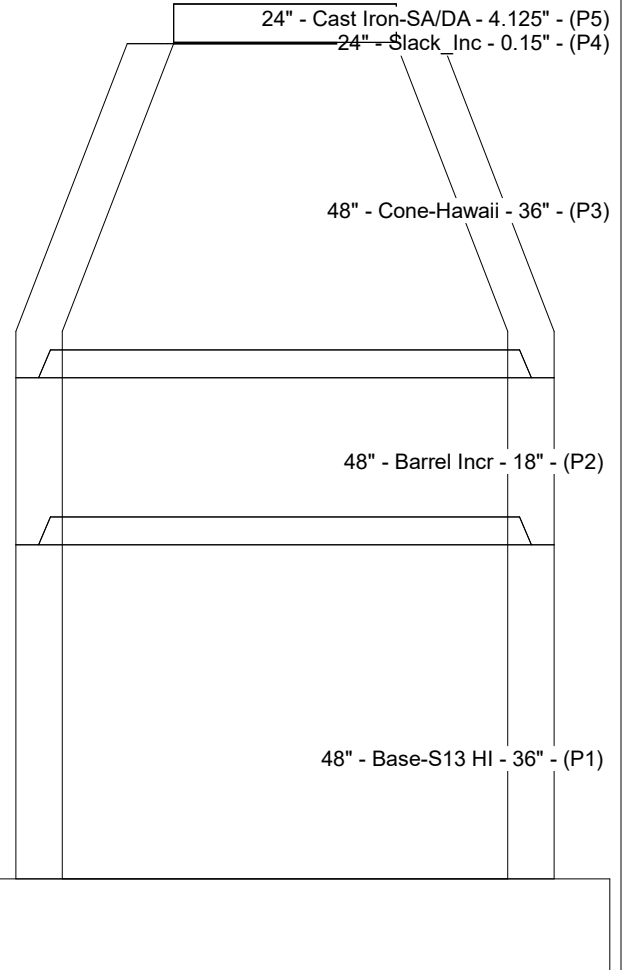
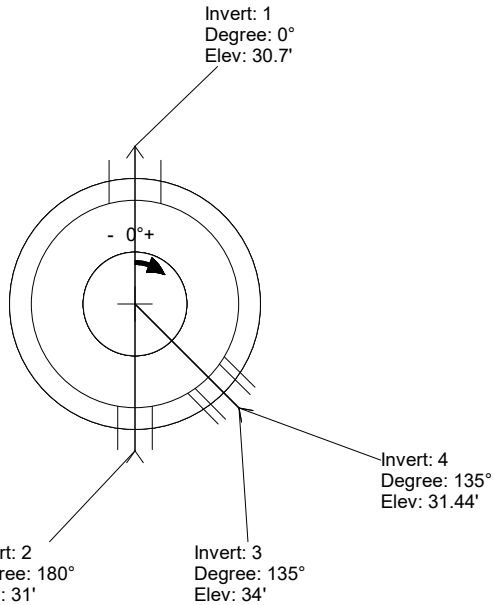
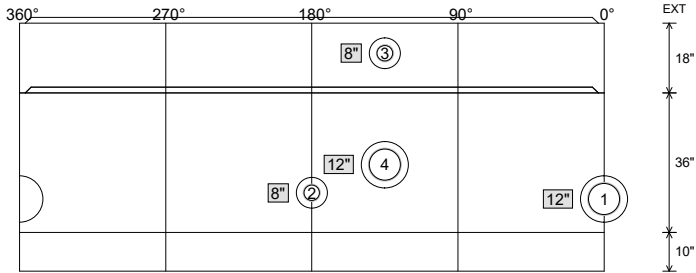
Slack: 0'
Floor (Top): 30.314'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

P5) 24" - Cast Iron-SA/DA - 4.125"
P4) 24" - Slack_Inc - 0.15"
P3) 48" - Cone-Hawaii - 36"
P2) 48" - Barrel Incr - 18"
P1) 48" - Base-S13 HI - 36"
2) 48" - 1" Joint Sealant - 1 Roll/Joint
2) 12" - Blockout - 12BO
2) 8" - Blockout - 8BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.17'						
Reducer								
Invert 1	P1	30.7'	0°	0"	0"	2.75"	12" Blockout 12BO	8" Pipe
Invert 2	P1	31'	180°	0"	75.5"	6.25"	8" Blockout 8BO	4" Pipe
Invert 3	P2	34'	135°	0"	56.5"	6.25"	8" Blockout 8BO	4" Pipe
Invert 4	P1	31.44'	135°	0"	56.5"	11.5"	12" Blockout 12BO	8" Pipe
Invert 5								
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

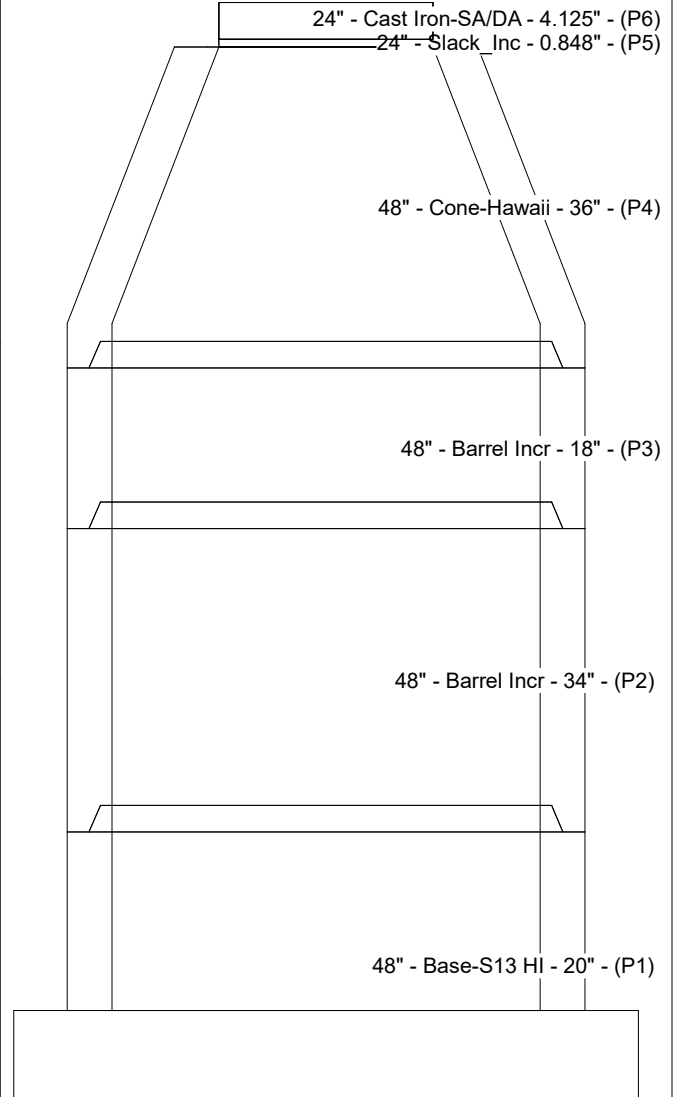
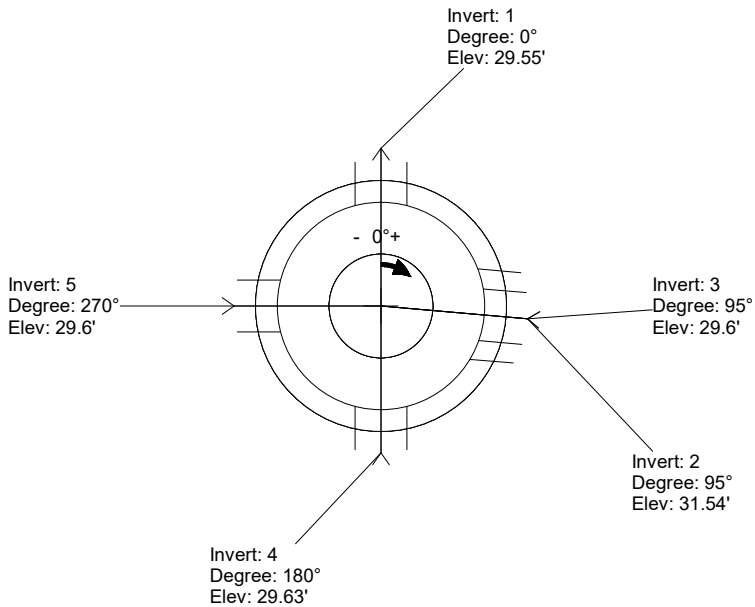
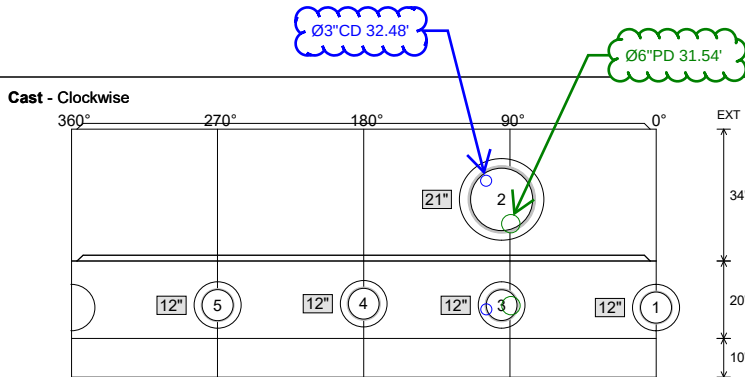
Structure ID: **SSMH-2**
Type: MH
Size: 48"
Rim: 38.63'
Invert: 29.55'
Rim to Invert: 9.08'

Slack: -0.002'
Floor (Top): 29.217'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P6) 24" - Cast Iron-SA/DA - 4.125"
P5) 24" - Slack_Inc - 0.848"
P4) 48" - Cone-Hawaii - 36"
P3) 48" - Barrel Incr - 18"
P2) 48" - Barrel Incr - 34"
P1) 48" - Base-S13 HI - 20"
3) 48" - 1" Joint Sealant - 1 Roll/Joint
4) 12" - Blockout - 12BO
1) 21" - Blockout - 21BO



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.632'						
Reducer								
Invert 1	P1	29.55'	0°	0"	0"	2"	12" Blockout 12BO	8" Pipe
Invert 2	P2	31.54'	95°	0"	39.75"	5.5"	21" Blockout 21BO	16" Pipe
Invert 3	P1	29.6'	95°	0"	39.75"	2.5"	12" Blockout 12BO	8" Pipe
Invert 4	P1	29.63'	180°	0"	75.5"	3"	12" Blockout 12BO	8" Pipe
Invert 5	P1	29.6'	270°	0"	113"	2.5"	12" Blockout 12BO	8" Pipe
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-3**
Type: MH
Size: 60"
Rim: 38.99'
Invert: 27.58'
Rim to Invert: 11.41'

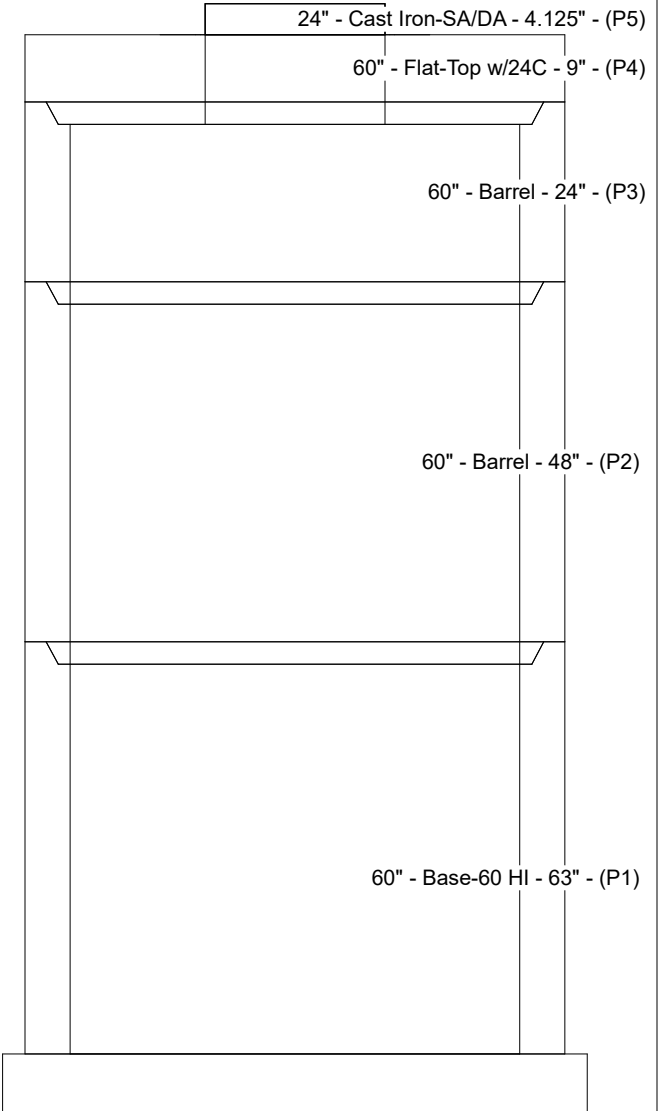
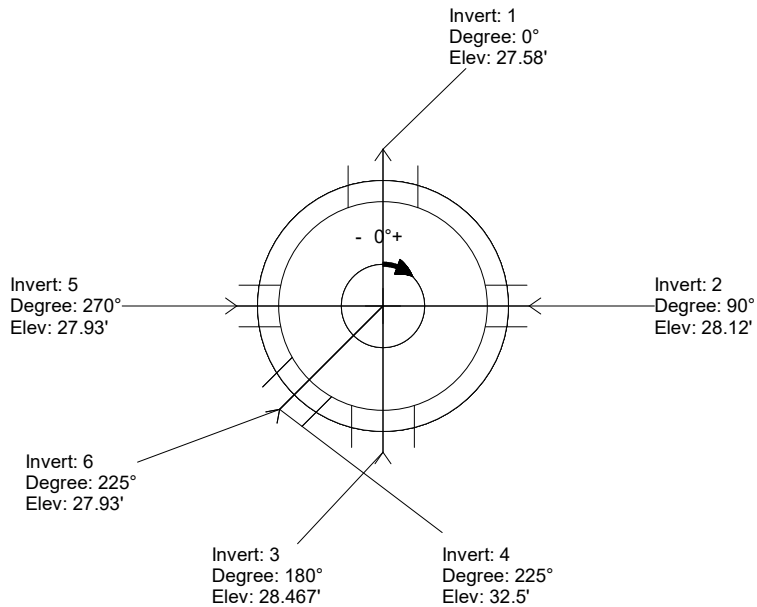
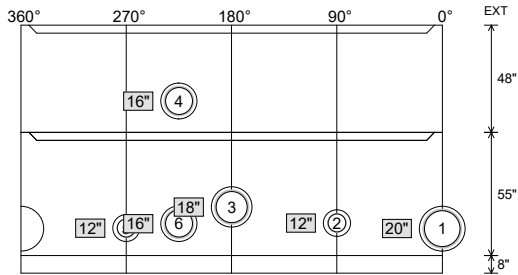
Slack: 0.066'
Floor (Top): 27.247'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P5) 24" - Cast Iron-SA/DA - 4.125"
P4) 60" - Flat-Top w/24C - 9"
P3) 60" - Barrel - 24"
P2) 60" - Barrel - 48"
P1) 60" - Base-60 HI - 63"
3) 60" - 1" Joint Sealant - 1.25 Rolls/Joint
1) 20" - Blockout - 20BO
2) 12" - Blockout - 12BO
1) 18" - Blockout - 18BO
2) 16" - Blockout - 16BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.924'						
Reducer								
Invert 1	P1	27.58'	0°	0"	0"	2"	20" Blockout 20BO	16" Pipe
Invert 2	P1	28.12'	90°	0"	47"	8.5"	12" Blockout 12BO	8" Pipe
Invert 3	P1	28.467'	180°	0"	94.25"	12.75"	18" Blockout 18BO	14" Pipe
Invert 4	P2	32.5'	225°	0"	117.75"	6"	16" Blockout 16BO	12" Pipe
Invert 5	P1	27.93'	270°	0"	141.25"	6.25"	12" Blockout 12BO	8" Pipe
Invert 6	P1	27.93'	225°	0"	117.75"	6.25"	16" Blockout 16BO	12" Pipe
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-4**
Type: MH
Size: 48"
Rim: 39.75'
Invert: 29.5'
Rim to Invert: 10.25'

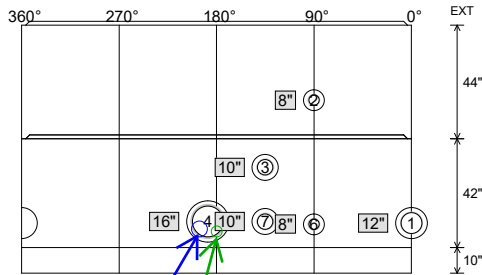
Slack: -0.002'
Floor (Top): 29.167'
Step Position:

Infrastructure
JENSEN

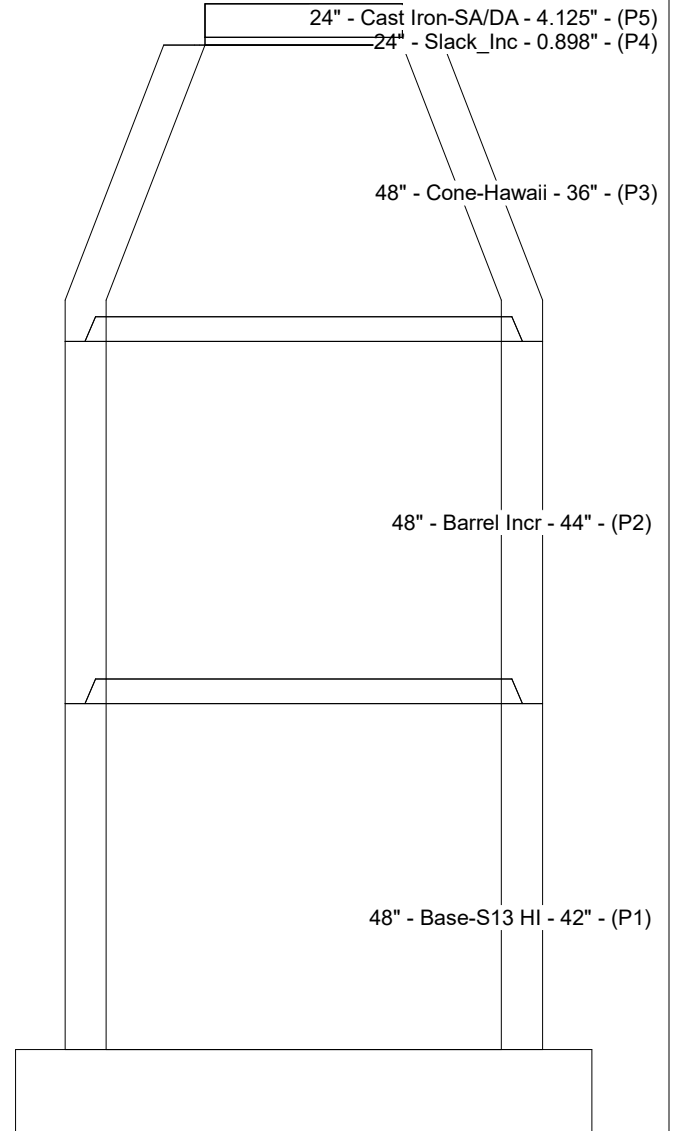
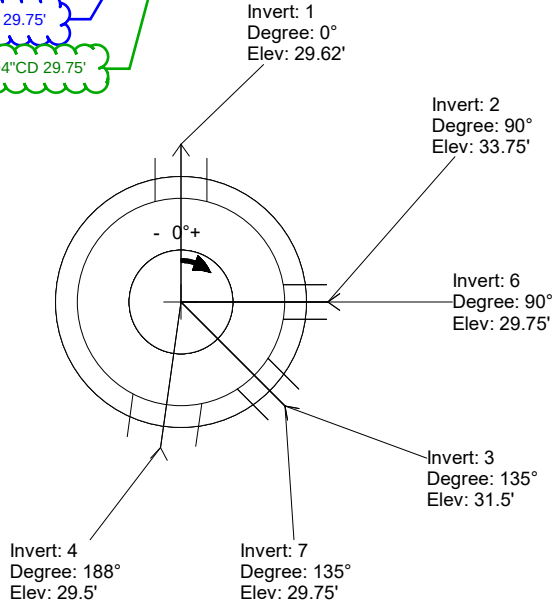
Quote ID: 2025090011

- P5) 24" - Cast Iron-SA/DA - 4.125"
P4) 24" - Slack_Inc - 0.898"
P3) 48" - Cone-Hawaii - 36"
P2) 48" - Barrel Incr - 44"
P1) 48" - Base-S13 HI - 42"
2) 48" - 1" Joint Sealant - 1 Roll/Joint
1) 12" - Blockout - 12BO
2) 8" - Blockout - 8BO
2) 10" - Blockout - 10BO
1) 16" - Blockout - 16BO

Cast - Clockwise



Ø6"PD 29.75'
Ø4"CD 29.75'



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		39.752'						
Reducer								
Invert 1	P1	29.62'	0°	0"	0"	3.5"	12" Blockout 12BO	8" Pipe
Invert 2	P2	33.75'	90°	0"	37.75"	11"	8" Blockout 8BO	4" Pipe
Invert 3	P1	31.5'	135°	0"	56.5"	26"	10" Blockout 10BO	6" Pipe
Invert 4	P1	29.5'	188°	0"	78.75"	2"	16" Blockout 16BO	12" Pipe
Invert 5								
Invert 6	P1	29.75'	90°	0"	37.75"	5"	8" Blockout 8BO	4" Pipe
Invert 7	P1	29.75'	135°	0"	56.5"	5"	10" Blockout 10BO	6" Pipe
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-5**

Type: MH

Size: 48"

Rim: 39'

Invert: 29'

Rim to Invert: 10'

Slack: -0.002'

Floor (Top): 28.667'

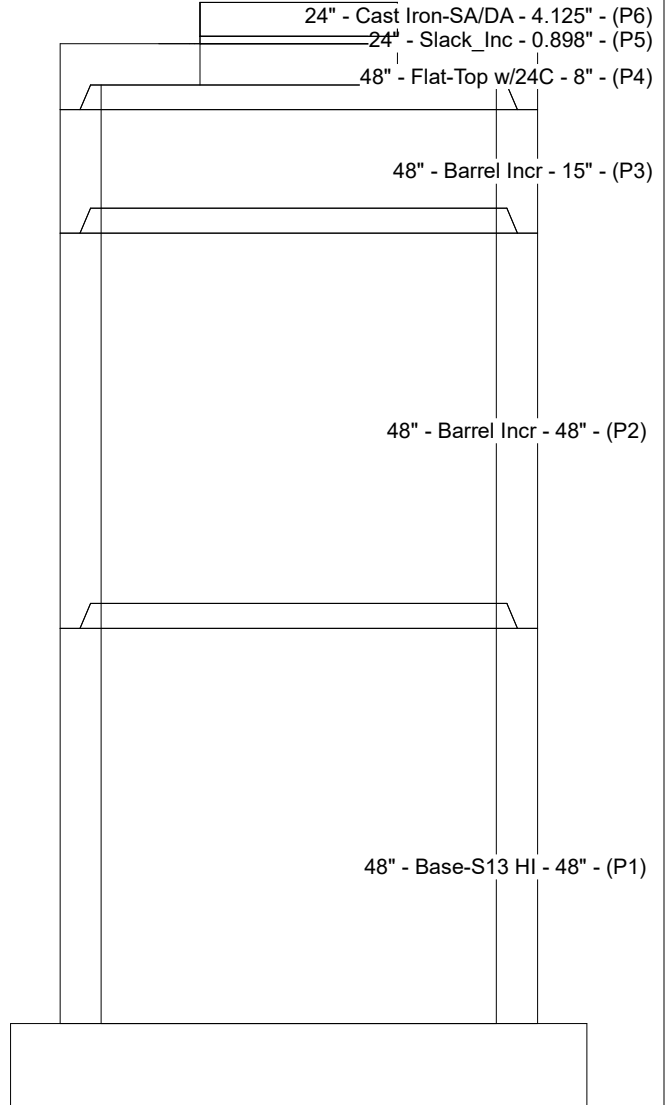
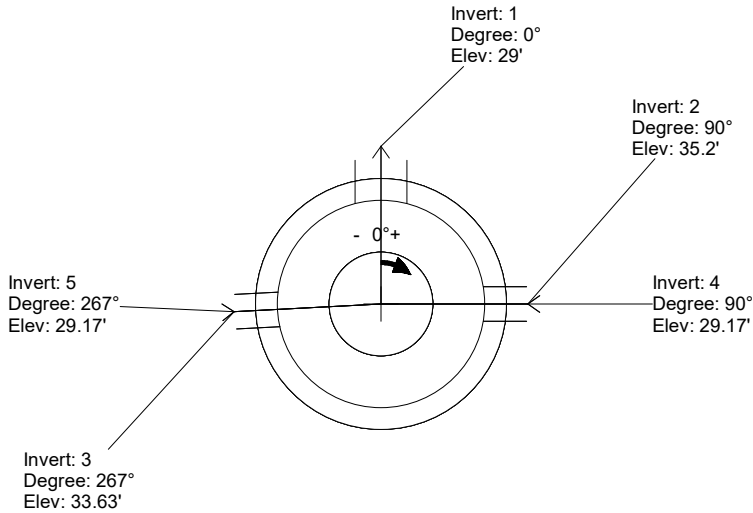
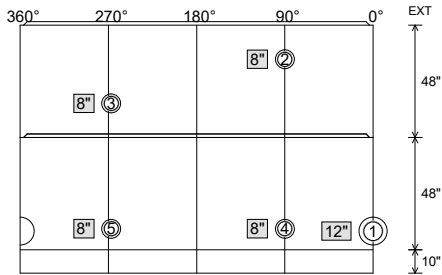
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P6) 24" - Cast Iron-SA/DA - 4.125"
P5) 24" - Slack_Inc - 0.898"
P4) 48" - Flat-Top w/24C - 8"
P3) 48" - Barrel Incr - 15"
P2) 48" - Barrel Incr - 48"
P1) 48" - Base-S13 HI - 48"
3) 48" - 1" Joint Sealant - 1 Roll/Joint
1) 12" - Blockout - 12BO
4) 8" - Blockout - 8BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		39.002'						
Reducer								
Invert 1	P1	29'	0°	0"	0"	2"	12" Blockout 12BO	8" Pipe
Invert 2	P2	35.2'	90°	0"	37.75"	29.5"	8" Blockout 8BO	6" Pipe
Invert 3	P2	33.63'	267°	0"	111.75"	10.5"	8" Blockout 8BO	6" Pipe
Invert 4	P1	29.17'	90°	0"	37.75"	5"	8" Blockout 8BO	6" Pipe
Invert 5	P1	29.17'	267°	0"	111.75"	5"	8" Blockout 8BO	6" Pipe
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-5A**
Type: MH
Size: 48"
Rim: 38.99'
Invert: 28.467'
Rim to Invert: 10.523'

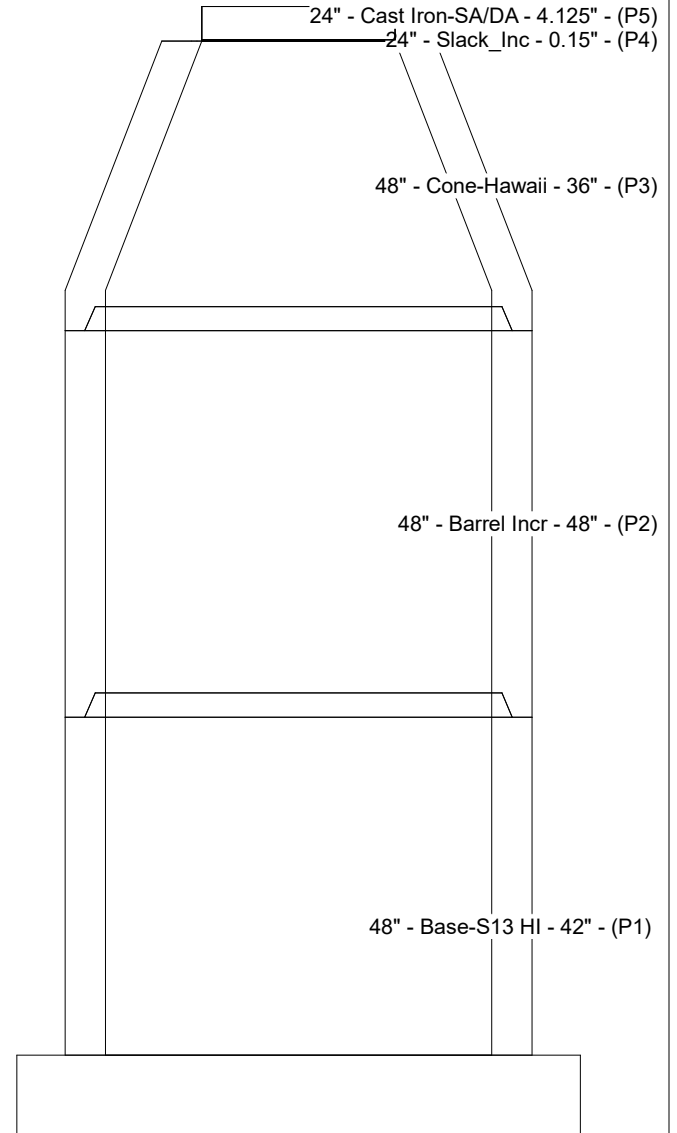
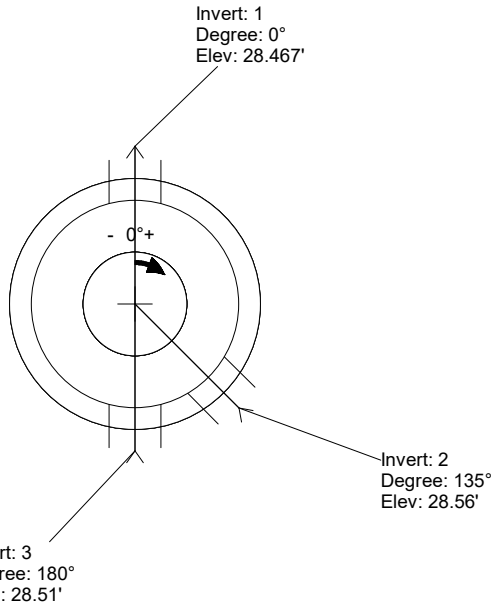
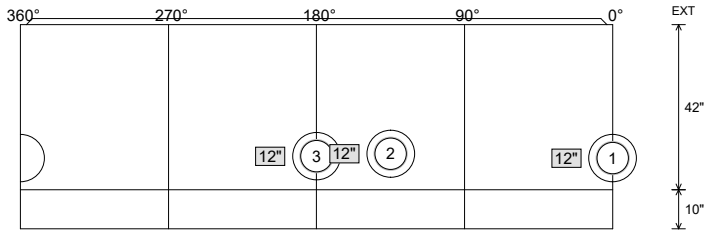
Slack: 0'
Floor (Top): 28.134'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P5) 24" - Cast Iron-SA/DA - 4.125"
P4) 24" - Slack_Inc - 0.15"
P3) 48" - Cone-Hawaii - 36"
P2) 48" - Barrel Incr - 48"
P1) 48" - Base-S13 HI - 42"
2) 48" - 1" Joint Sealant - 1 Roll/Joint
3) 12" - Blockout - 12BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.99'						
Reducer								
Invert 1	P1	28.467'	0°	0"	0"	2"	12" Blockout 12BO	8" Pipe
Invert 2	P1	28.56'	135°	0"	56.5"	3"	12" Blockout 12BO	8" Pipe
Invert 3	P1	28.51'	180°	0"	75.5"	2.5"	12" Blockout 12BO	8" Pipe
Invert 4								
Invert 5								
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-6**
Type: MH
Size: 48"
Rim: 38.4'
Invert: 30.74'
Rim to Invert: 7.66'

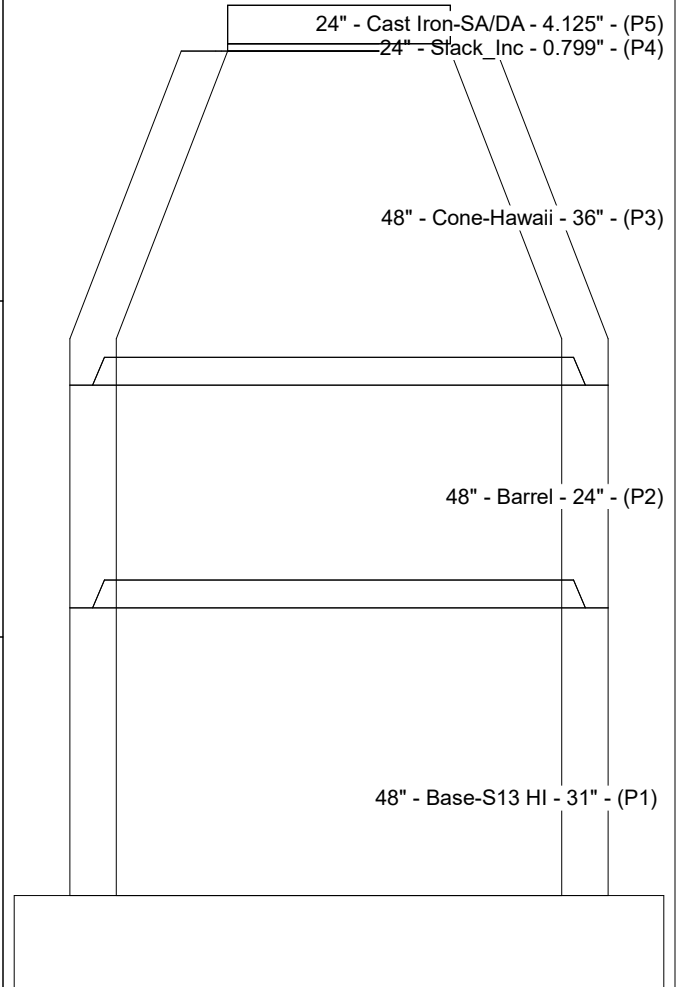
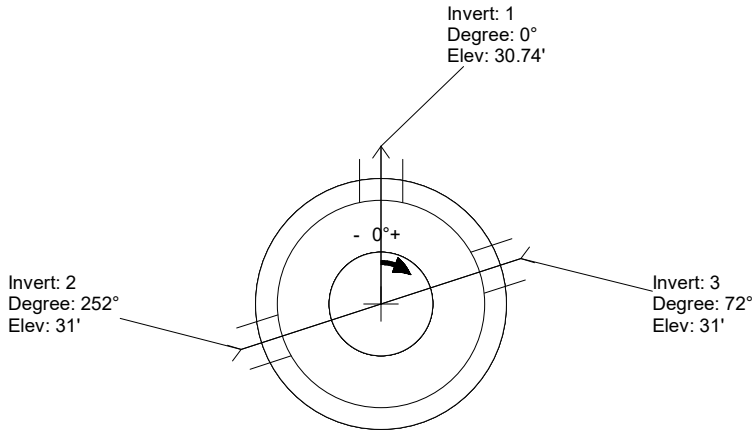
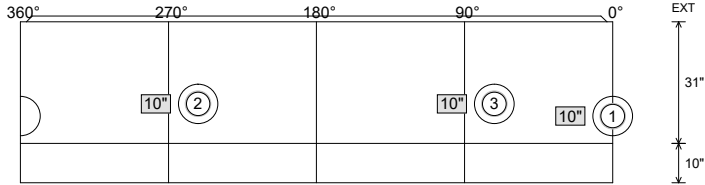
Slack: -0.001'
Floor (Top): 30.407'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P5) 24" - Cast Iron-SA/DA - 4.125"
P4) 24" - Slack_Inc - 0.799"
P3) 48" - Cone-Hawaii - 36"
P2) 48" - Barrel - 24"
P1) 48" - Base-S13 HI - 31"
2) 48" - 1" Joint Sealant - 1 Roll/Joint
3) 10" - Blockout - 10BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.401'						
Reducer								
Invert 1	P1	30.74'	0°	0"	0"	2"	10" Blockout 10BO	6" Pipe
Invert 2	P1	31'	252°	0"	105.5"	5"	10" Blockout 10BO	6" Pipe
Invert 3	P1	31'	72°	0"	30.25"	5"	10" Blockout 10BO	6" Pipe
Invert 4								
Invert 5								
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-6A**

Type: MH

Size: 48"

Rim: 38.5'

Invert: 32.72'

Rim to Invert: 5.78'

Slack: 0.019'

Floor (Top): 32.387'

Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

P3) 24" - Cast Iron-SA/DA - 4.125"

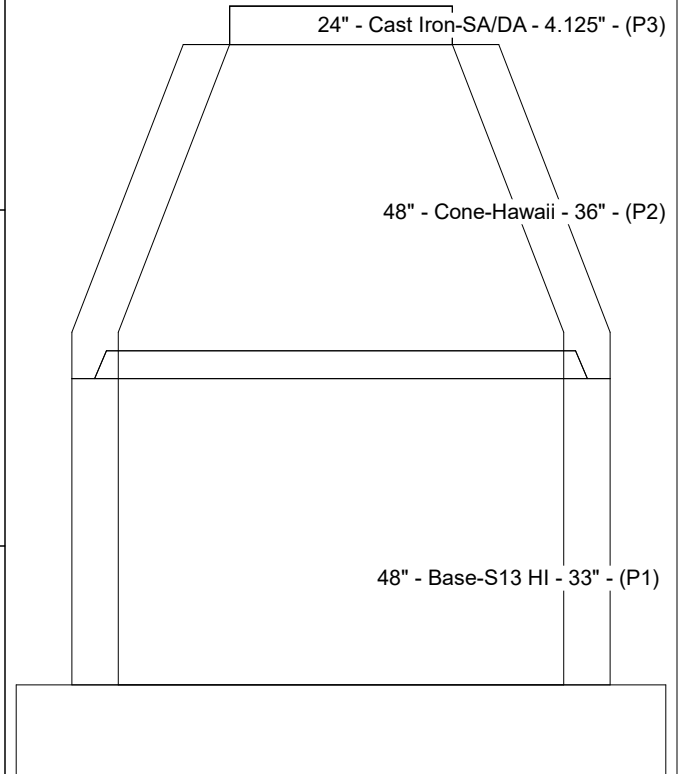
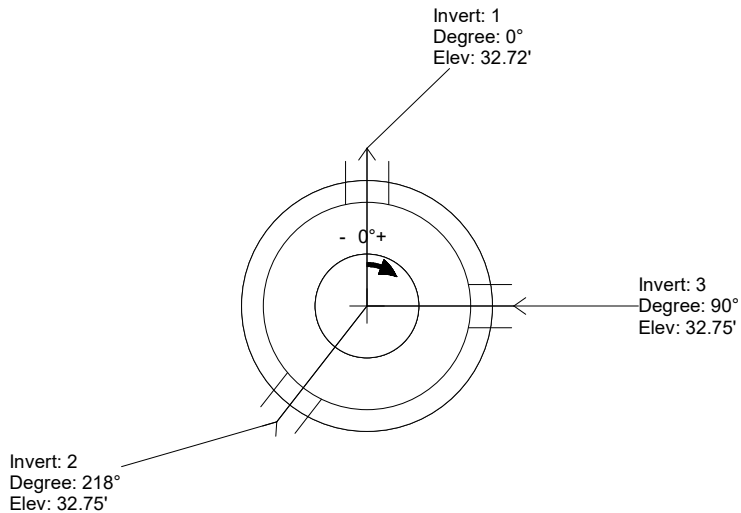
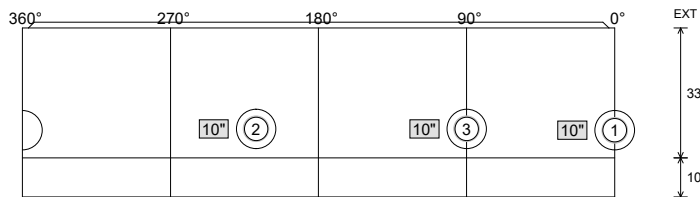
P2) 48" - Cone-Hawaii - 36"

P1) 48" - Base-S13 HI - 33"

1) 48" - 1" Joint Sealant - 1 Roll/Joint

3) 10" - Blockout - 10BO

Cast - Clockwise



Structure Notes:

Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.481'						
Reducer								
Invert 1	P1	32.72'	0°	0"	0"	2"	10" Blockout 10BO	6" Pipe
Invert 2	P1	32.75'	218°	0"	91.25"	2.25"	10" Blockout 10BO	6" Pipe
Invert 3	P1	32.75'	90°	0"	37.75"	2.25"	10" Blockout 10BO	6" Pipe
Invert 4								
Invert 5								
Invert 6								
Invert 7								
Invert 8								

Jensen Infrastructure
Hilo WWTP
Nan Inc
Hilo, HI
Jorelle Flint

Structure ID: **SSMH-7**
Type: MH
Size: 48"
Rim: 38.4'
Invert: 34.664'
Rim to Invert: 3.736'

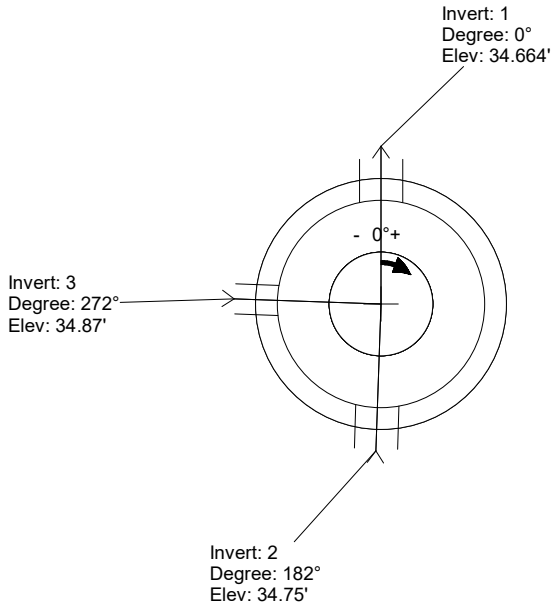
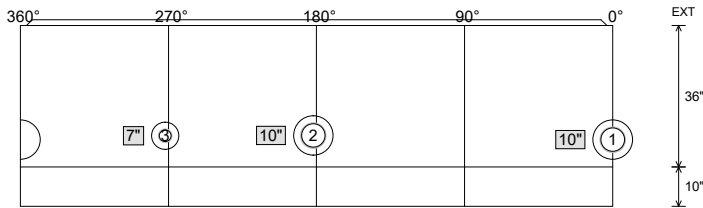
Slack: 0.05'
Floor (Top): 34.331'
Step Position:

Infrastructure
JENSEN

Quote ID: 2025090011

- P4) 24" - Cast Iron-SA/DA - 4.125"
P3) 24" - Slack_Inc - 0.1"
P2) 48" - Flat-Top w/24C - 8"
P1) 48" - Base-S13 HI - 36"
1) 48" - 1" Joint Sealant - 1 Roll/Joint
2) 10" - Blockout - 10BO
1) 7" - Blockout - 7BO

Cast - Clockwise



24" - Cast Iron-SA/DA - 4.125" - (P4)
24" - Slack_Inc - 0.1" - (P3)
48" - Flat-Top w/24C - 8" - (P2)

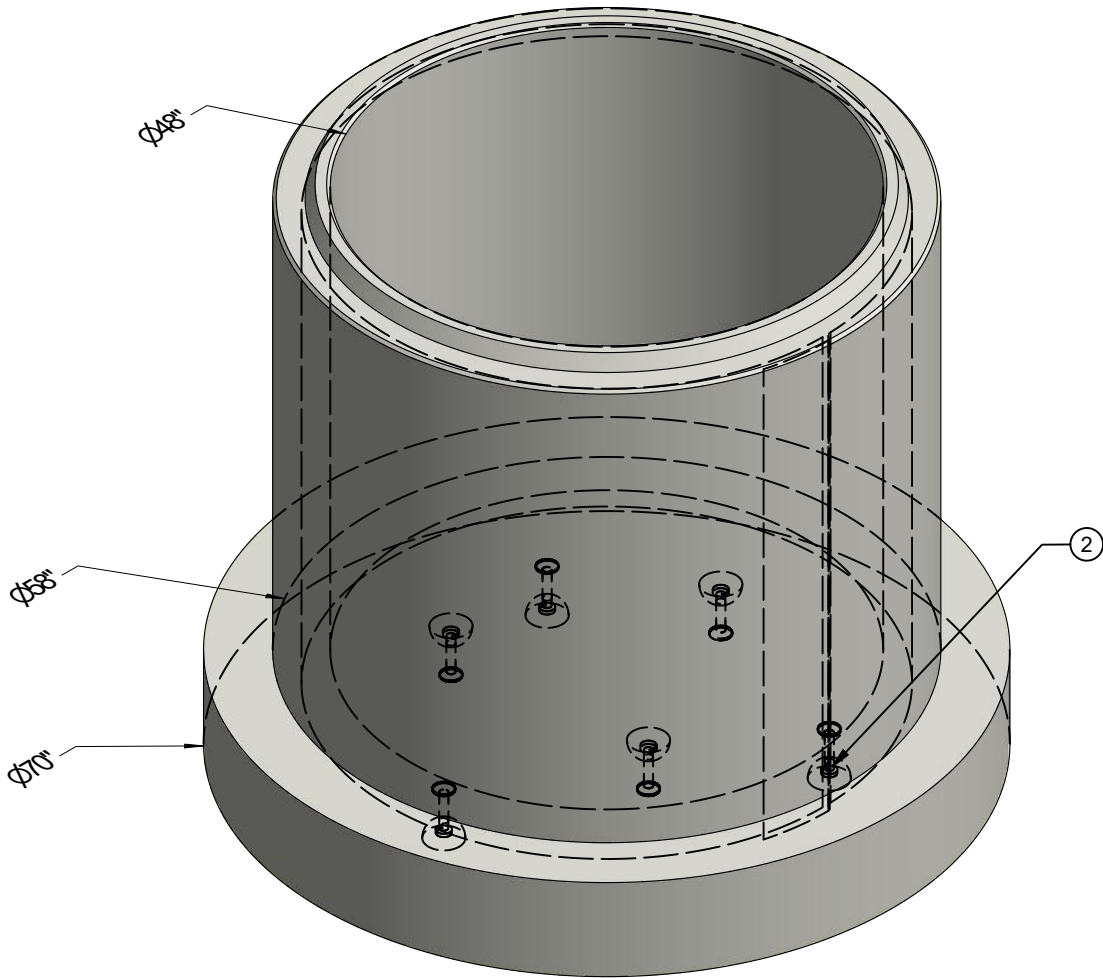
48" - Base-S13 HI - 36" - (P1)

Structure Notes:

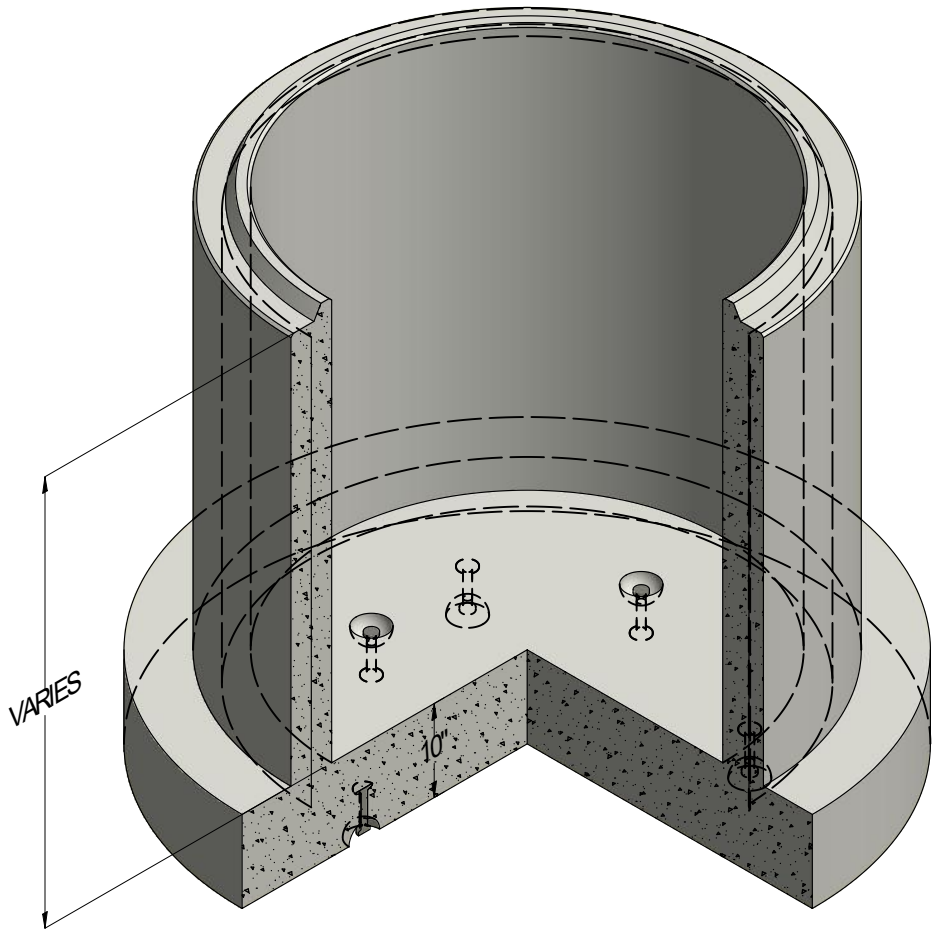
Position	Ref	Elev	Angle	Offset	Int. (cw)	Up ()	Connector	Pipe
Rim		38.35'						
Reducer								
Invert 1	P1	34.664'	0°	0"	0"	2"	10" Blockout 10BO	6" Pipe
Invert 2	P1	34.75'	182°	0"	76.25"	3"	10" Blockout 10BO	6" Pipe
Invert 3	P1	34.87'	272°	0"	114"	4.5"	7" Blockout 7BO	3" Pipe
Invert 4								
Invert 5								
Invert 6								
Invert 7								
Invert 8								

NOTES
■ FOR COMPLETE DESIGN AND PRODUCT INFORMATION CONTACT JENSEN PRECAST.

BILL OF MATERIALS				
ITEM	UNIT QTY	QTY	PART NUMBER	DESCRIPTION
1	45.568 ft^3	45.568 ft^3	100011428	CONCRETE MIX GENERIC 6500 PSI
2	1	6	100002874	ANCHOR DOGBONE P52 4T X 4-3/4" GALV
3	1	1	300122360	REINFORCING MAT 300122360
4	60.400 ft^2	60.400 ft^2	100015040	REINF WWR MH 3X8 W3.75xw2 96IN ASTM A-1024



ISO REF



SECTION REF.

LIFTING NOTES			
■ THIS CONCRETE PRODUCT IS DESIGNED TO BE LIFTED AT 2000 PSI WITH 3 LIFT CONNECTIONS EQUALLY LOADED TO 3 LIFTING POINTS AT A MINIMUM 60° SLING ANGLE, MEASURED FROM HORIZONTAL.			
BURY DEPTH:		5.0' - 9.0' to Inside Floor	
WATER LEVEL:	3.0'	WEIGHT:	VARIES
DESIGN SPEC:	ASTM C478,AASHTO LRFD	LOADING SPEC:	Aircraft
REINFORCEMENT TYPE:	Rebar,WWR Plain	LIFTING TYPE:	Dogbone

JENSEN
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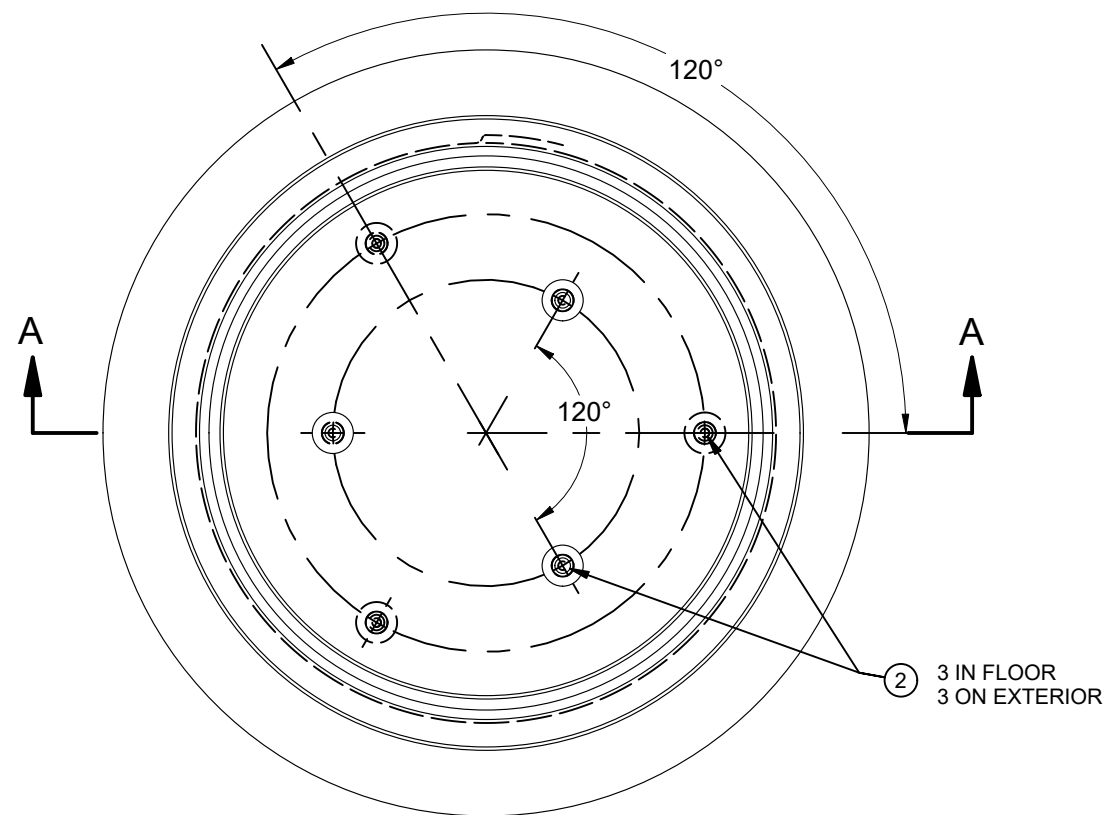


EST. 1968

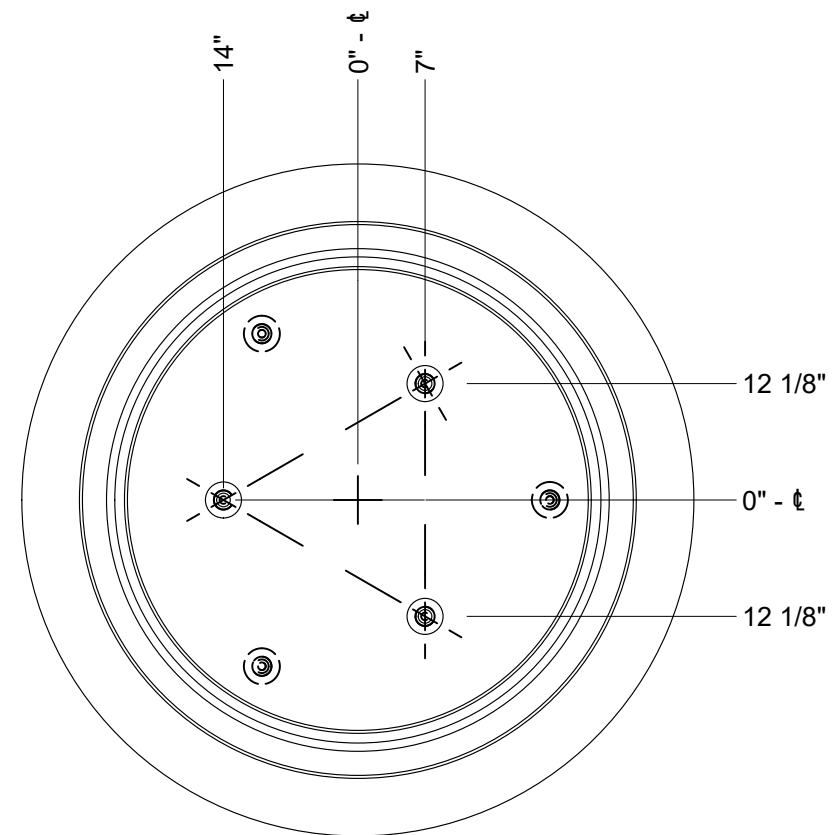
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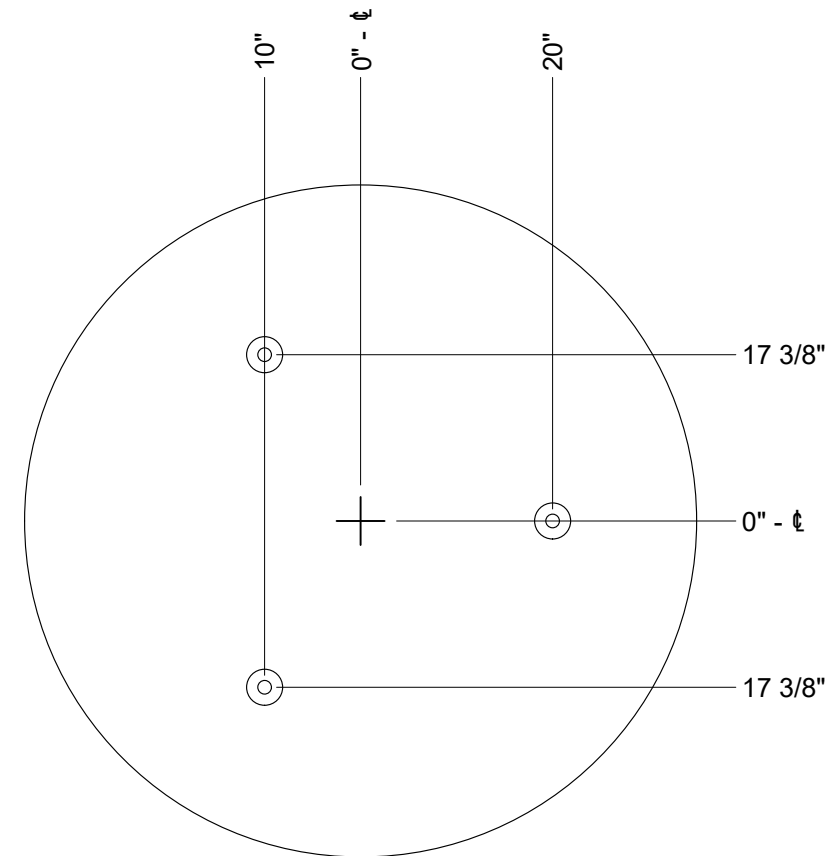
REVISION HISTORY			
REV	DESCRIPTION	DATE	BY
A	1909-18780-HAW	11-20-19	ACHO
B	WIRE MESH REINFORCING ASSEMBLY REMOVED	11-16-20	ACHO
DESCRIPTION:			REV:
Manhole D4858 Rnd. Bottom Section TU Manually Configured			B-2
CASTING LAYOUT			
PART NUMBER:		DRAWN BY:	
200031089		ACHO	
CREATED:		MODIFIED:	
5/28/2019		11/17/2020	
			SHEET:
			1 OF 3



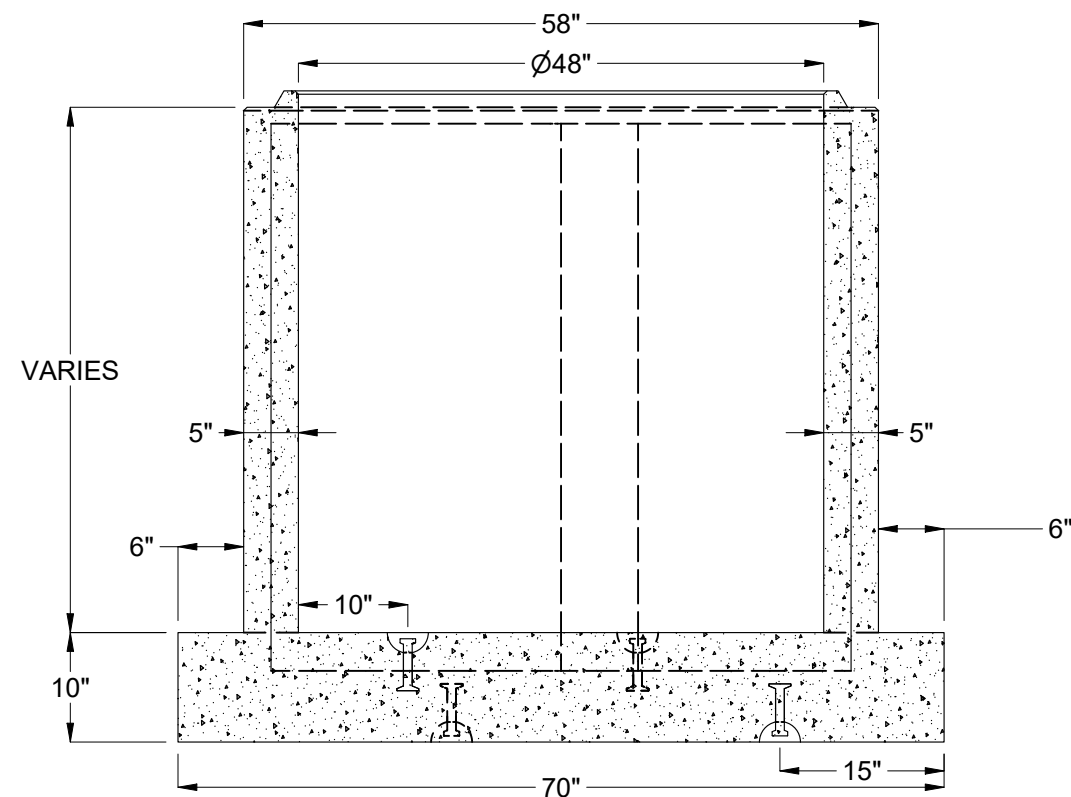
PLAN VIEW



PLAN VIEW
INSIDE FLOOR



PLAN VIEW
OUTSIDE OF FLOOR



SECTION A-A

BURY DEPTH:		5.0' - 9.0' to Inside Floor	
WATER LEVEL:	3.0'	WEIGHT:	VARIES
DESIGN SPEC:	ASTM C478,AASHTO LRFD	LOADING SPEC:	Aircraft
REINFORCEMENT TYPE:	Rebar,WWR Plain	LIFTING TYPE:	Dogbone

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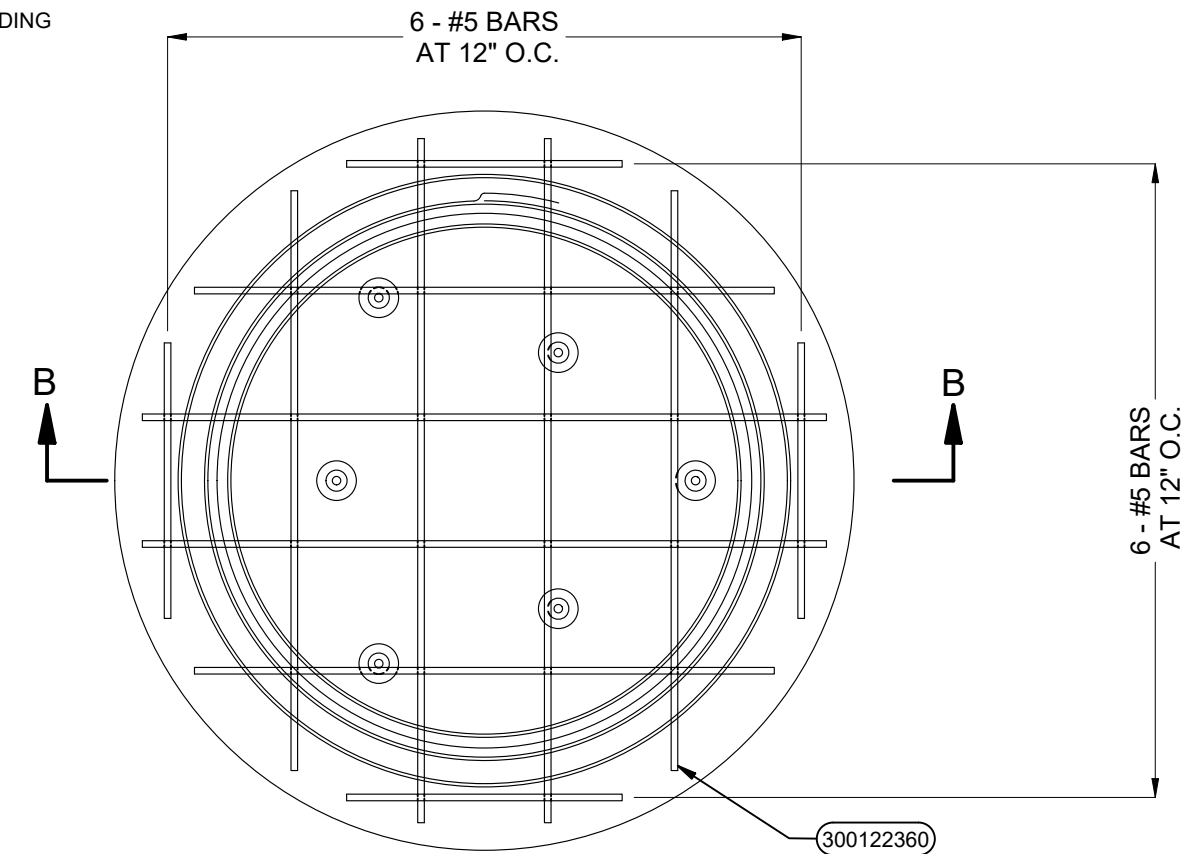
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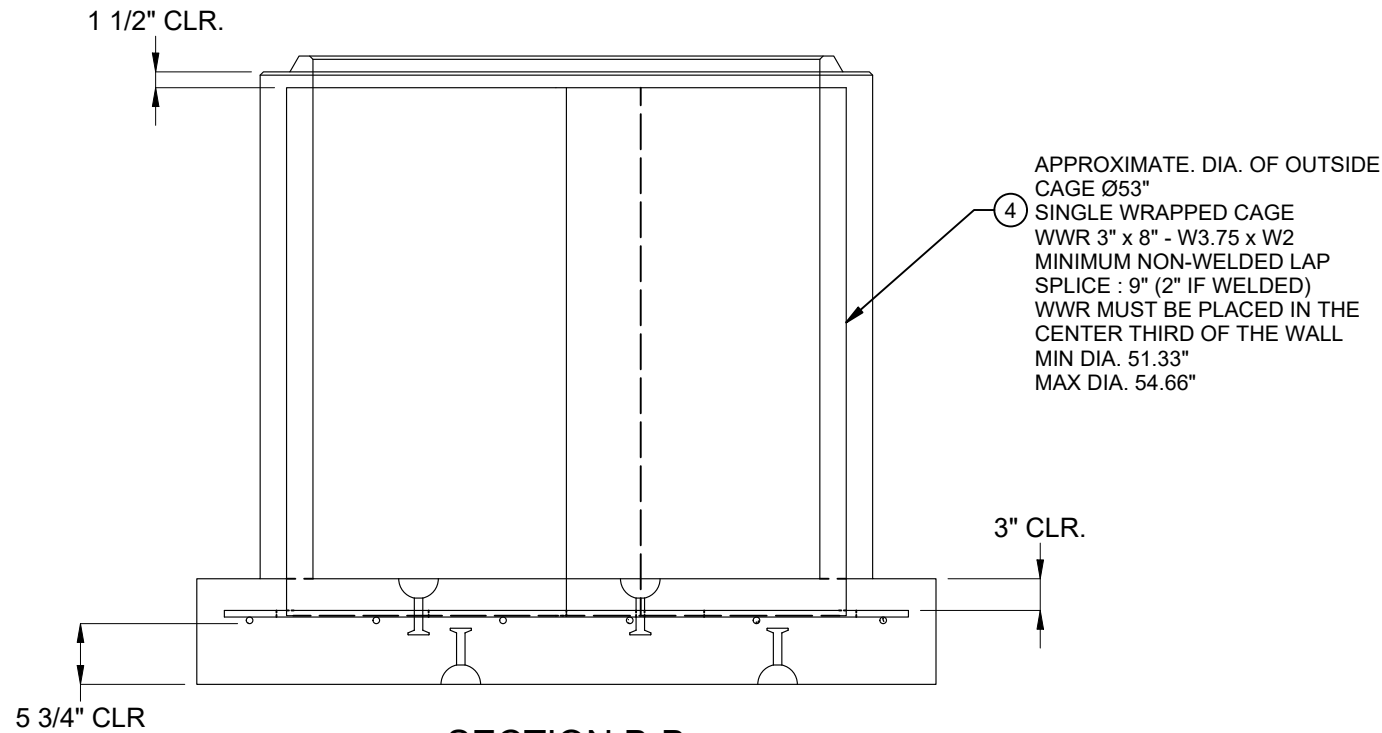
DESCRIPTION:			
Manhole D4858 Rnd. Bottom Section TU Manually Configured			
CASTING DRAWING			
PART NUMBER:	200031089	DRAWN BY:	ACHO
CREATED:	5/28/2019	MODIFIED:	11/17/2020

REV:	B-2
SHEET:	2 OF 3

- ALL CONCRETE TO HAVE A 28 DAY COMPRESSIVE STRENGTH OF 6500 psi.
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARD.
- DEFORMED WIRE MESH SHALL COMPLY WITH ASTM A-1064 GRADE 80 STANDARDS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- BURY DEPTH 5.0' - 9.0' TO Inside Floor
- WATER LEVEL: 3.0'
- DESIGN SPECIFICATION: ASTM C478,AASHTO LRFD
- LOADING SPECIFICATION: Aircraft
- EQUIVALENT SOIL WEIGHT DRY - 120 lbs/cu.ft.
- EQUIVALENT LATERAL PRESSURE DRY - 40 lbs/cu.ft.
- EQUIVALENT SOIL WEIGHT WET - 120 lbs/cu.ft.
- EQUIVALENT LATERAL PRESSURE WET - 80 lbs/cu.ft.
- SOIL INTERACTION FACTOR : 1.15



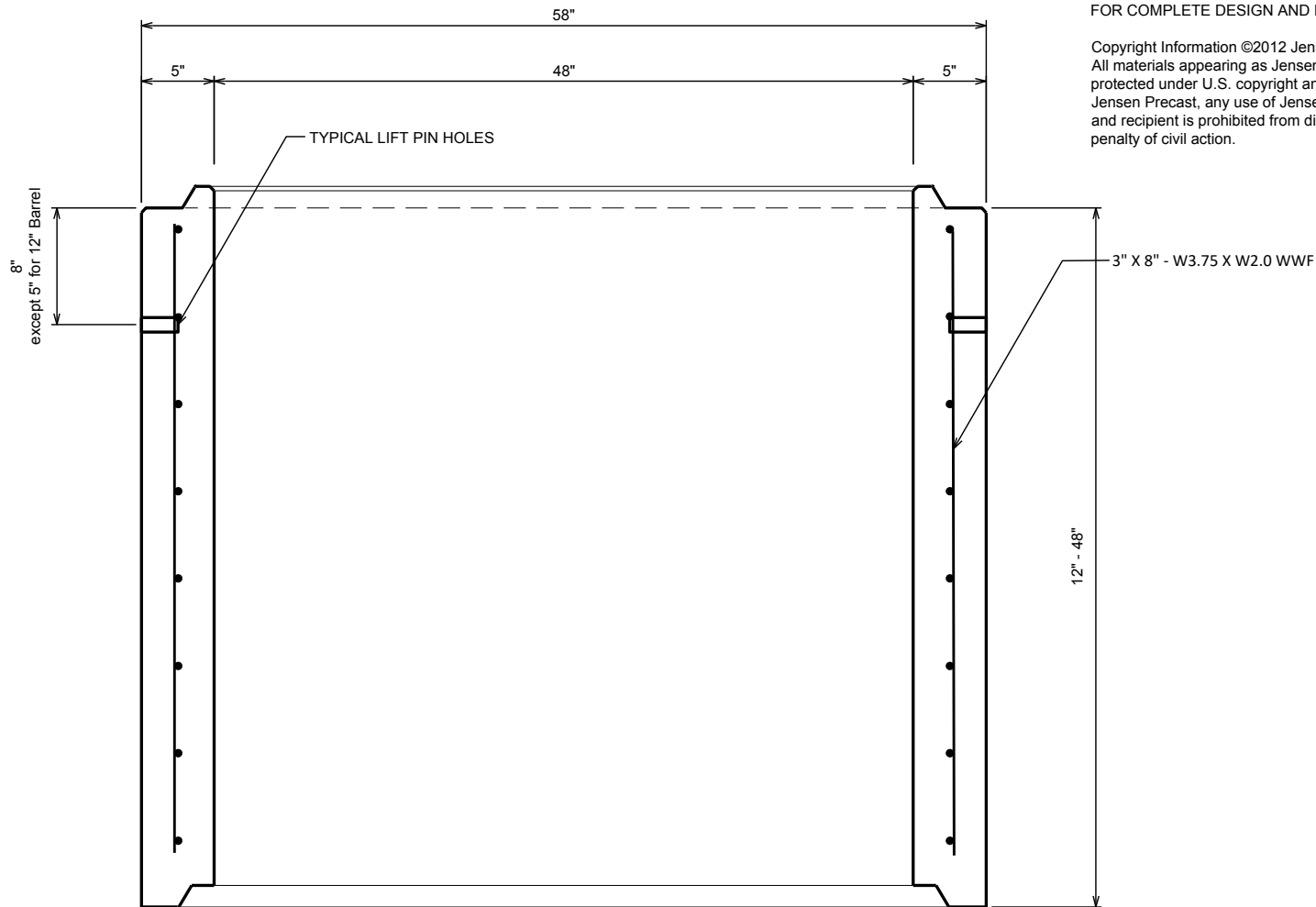
PLAN VIEW



SECTION B-B

BURY DEPTH:		5.0' - 9.0' to Inside Floor		JENSEN® PRECAST EST. 1968 WWW.JENSENPRECAST.COM	©2019 Jensen Precast- All rights reserved. All materials appearing as Jensen Precast documents and the like are proprietary work product and are protected under U.S. copyright and other laws. Unless in conjunction with business conducted with Jensen Precast, any use of Jensen Precast work product without express, written consent is prohibited, and recipient is prohibited from distributing any and all work product to non-approved third parties under penalty of civil action.	DESCRIPTION:			REV:	
WATER LEVEL:		3.0'	WEIGHT:			VARIES			B-2	
DESIGN SPEC:		ASTM C478,AASHTO LRFD	LOADING SPEC:			Aircraft			SHEET:	
REINFORCEMENT TYPE:		Rebar,WWR Plain	LIFTING TYPE:			Dogbone			3 OF 3	
						REINFORCING CAGE REVIEW				
						PART NUMBER:	200031089	DRAWN BY:	ACHO	
						CREATED:	5/28/2019	MODIFIED:	11/17/2020	

CHECKED BY: _____ DATE: _____



BARREL SECTION VIEW

Ø48" - 5" WALL
BARREL
ASTM C-478

JENSEN PRECAST

DESIGN NOTES:
CONFORMS TO ASTM C-478 AND AASHTO M-199
MINIMUM CONCRETE STRENGTH 5500 P.S.I.
CEMENT ASTM C-150
FLYASH ASTM C-618
AGGREGATES ASTM C-33
REINFORCEMENT SHALL MEET ASTM A185 OR A497

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825 STENERI WAY
SPARKS, NEVADA 89431
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3		
2		
1		
0	02-10-12	NEW DRAWING
REV.	DATE	REVISION

FILE NO: 4858X12-48 BARREL WWF REIN.DWG

NOTES
■ FOR COMPLETE DESIGN AND PRODUCT INFORMATION CONTACT JENSEN PRECAST.

BILL OF MATERIALS				
ITEM	UNIT QTY	QTY	PART NUMBER	DESCRIPTION
1	13.900 ft^3	13.900 ft^3	100002728	CONCRETE MIX GENERIC 4000 PSI
2	106.161 ft	106.161 ft	100007262	REINF HOOP STEEL .25235DIA GA ASTM A-1064 W4.7
3	8.509 ft	8.509 ft	100007262	REINF HOOP STEEL .25235DIA GA ASTM A-1064 W4.7
4	493.500 in	493.500 in	100003075	REINF #4 REBAR ASTM A706/A615 GR60



ISOMETRIC VIEW

LIFTING NOTES		
■ THIS CONCRETE PRODUCT IS DESIGNED TO BE LIFTED AT 2500PSI WITH 2 LIFT CONNECTIONS EQUALLY LOADED TO 2 LIFTING POINTS AT A SLING ANGLE BETWEEN 60° AND 90°, MEASURED FROM HORIZONTAL.		
BURY DEPTH:	0.0' - 0.3' Ft. to Top of Top Slab	
WATER LEVEL:	3.0' Ft.	WEIGHT: 2082.97 Lbs
DESIGN SPEC:	ASTM C478	LOADING SPEC: N/A
REINFORCEMENT TYPE:	Hoop Steel	LIFTING TYPE: Insert



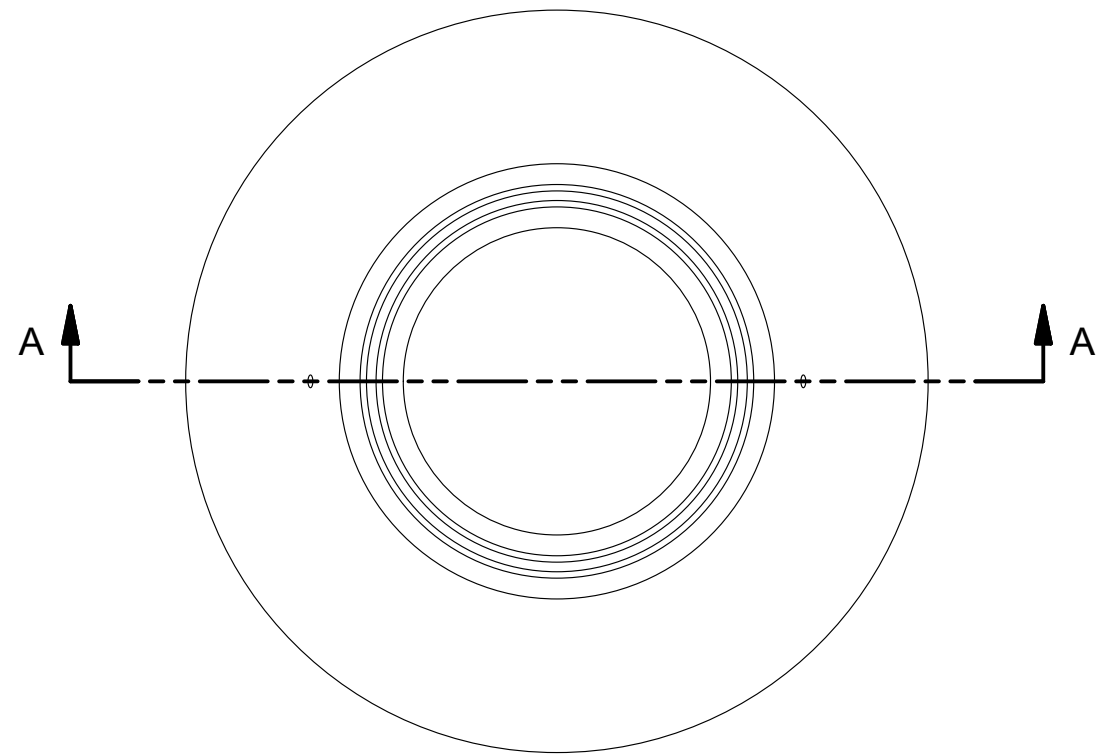
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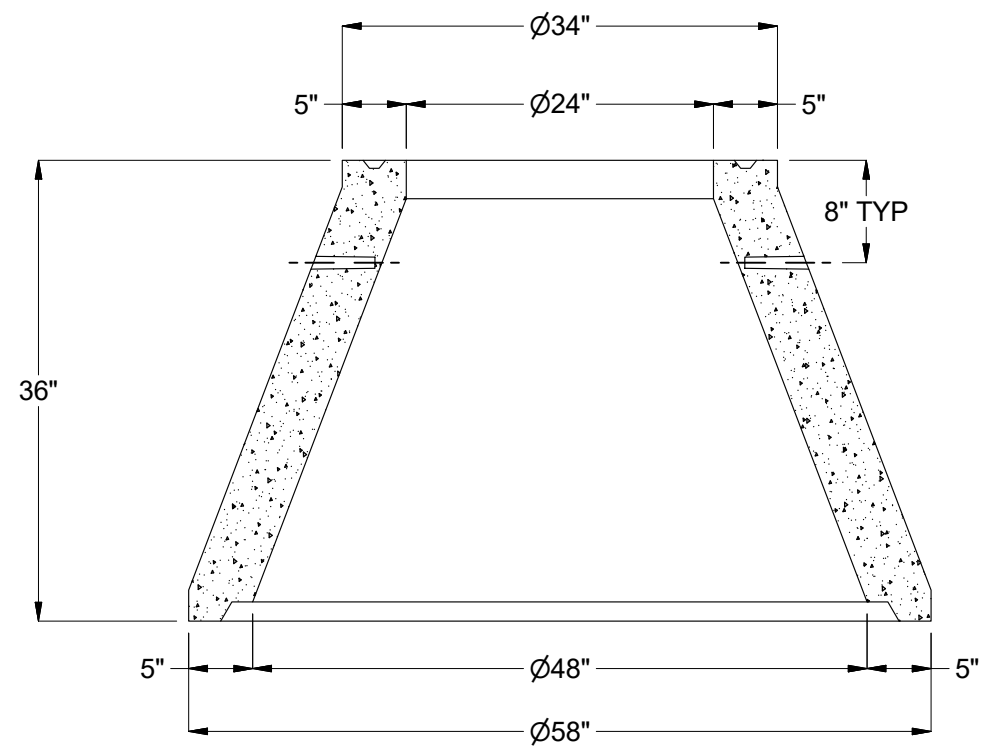
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REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	Initial Design	1/25/2019	
DESCRIPTION:			REV:
Manhole 4824 x Concentric			A
CASTING LAYOUT			
PART NUMBER:	200015036	DRAWN BY:	SCD
CREATED:	1/7/2018	MODIFIED:	1/25/2019
			SHEET:
			1 OF 3

200015036.dwg



PLAN VIEW



SECTION A-A



ISOMETRIC VIEW

BURY DEPTH:		0.0' - 0.3' Ft. to Top of Top Slab	
WATER LEVEL:	3.0' Ft.	WEIGHT:	2082.97 Lbs
DESIGN SPEC:	ASTM C478	LOADING SPEC:	N/A
REINFORCEMENT TYPE:	Hoop Steel	LIFTING TYPE:	Insert

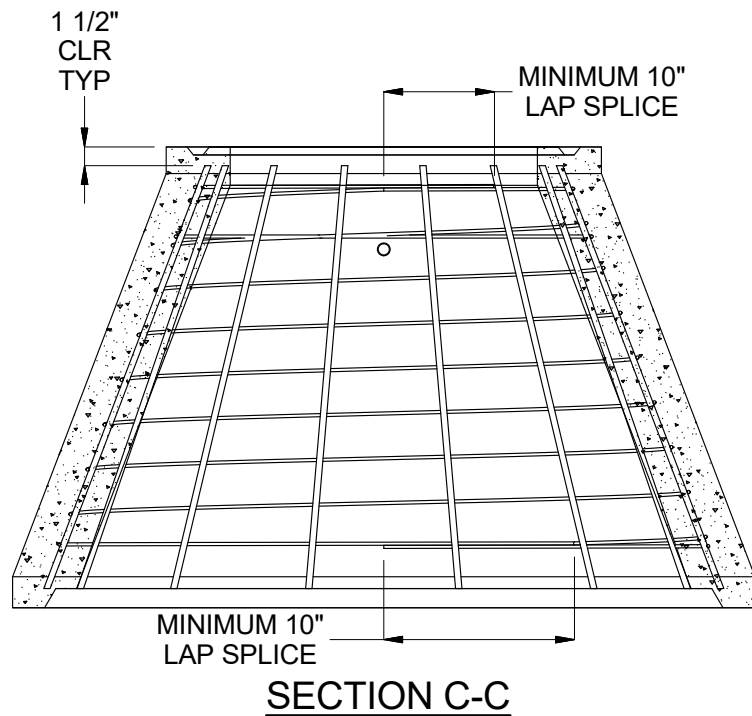
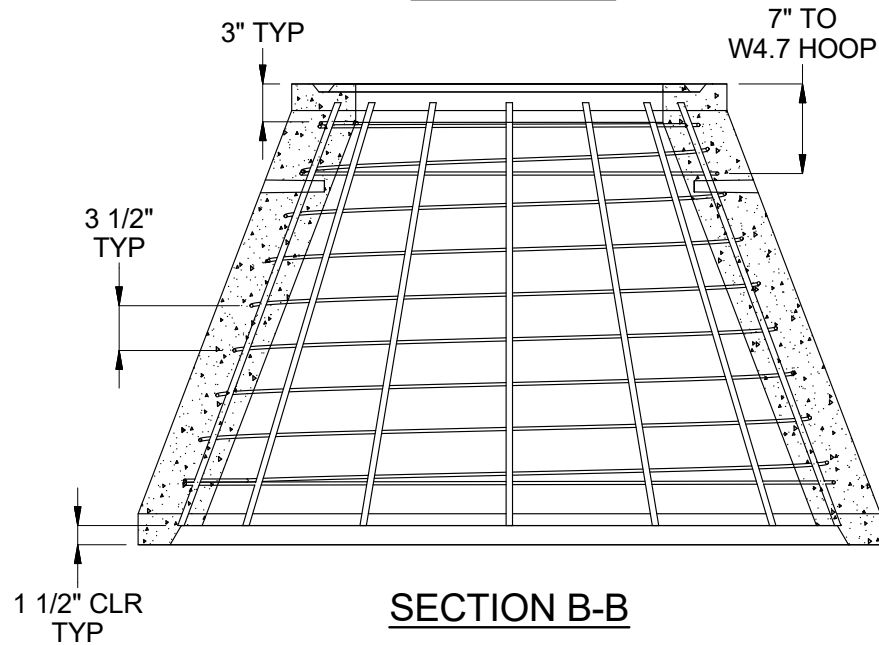
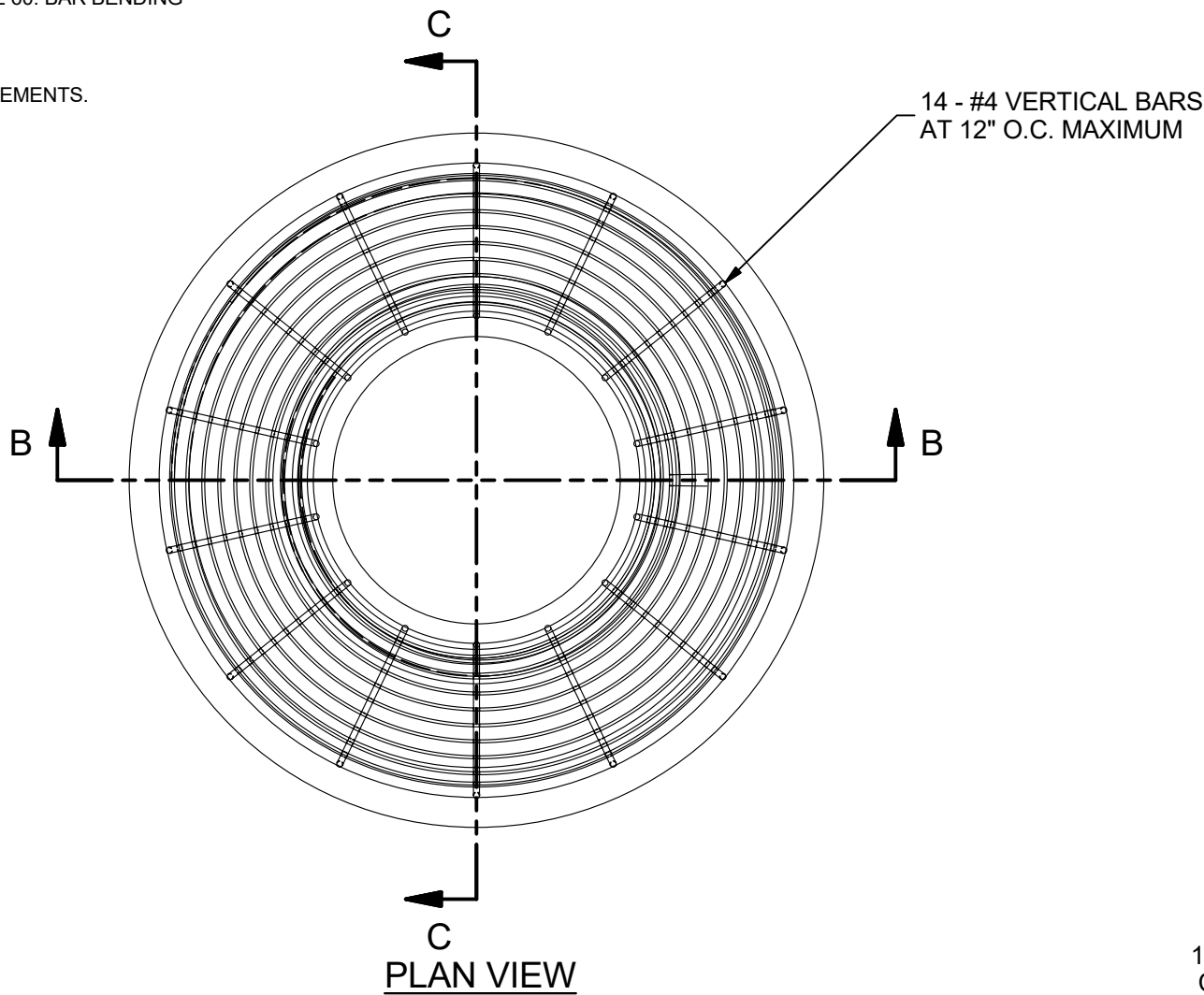


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DESCRIPTION:		Manhole 4824 x Concentric	
		CASTING DRAWING	
PART NUMBER:	200015036	DRAWN BY:	SCD
CREATED:	1/7/2018	MODIFIED:	1/25/2019

REV:	A
SHEET:	2 OF 3

- ALL CONCRETE TO HAVE A 28 DAY COMPRESSIVE STRENGTH OF 4000 psi.
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARD.
- DEFORMED WIRE MESH SHALL COMPLY WITH ASTM A-1064 GRADE 80 STANDARDS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- BURY DEPTH 0.0' - 0.3' TO Top of Top Slab
- WATER LEVEL: 3.0'
- DESIGN SPECIFICATION: ASTM C478
- LOADING SPECIFICATION: N/A



BURY DEPTH:		0.0' - 0.3' Ft. to Top of Top Slab	
WATER LEVEL:	3.0' Ft.	WEIGHT:	2082.97 Lbs
DESIGN SPEC:	ASTM C478	LOADING SPEC:	N/A
REINFORCEMENT TYPE:	Hoop Steel	LIFTING TYPE:	Insert

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		Manhole 4824 x Concentric		A	
		REINFORCING CAGE REVIEW		SHEET:	
		PART NUMBER:	200015036	DRAWN BY:	SCD
		CREATED:	1/7/2018	MODIFIED:	1/25/2019
				3 OF 3	



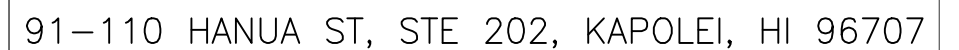
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- ALL CONCRETE TO HAVE 28 DAY COMPRESSIVE STRENGTH OF 5000 psi
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARDS.
- EQUIVALENT AREA $F_y=65000$ psi ASTM A 185 OR A 497 REINFORCING CAGES MY BE UTILIZED IN LIEU OF GRADE 60 REINFORCING BARS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- STRUCTURAL DESIGN IS BASED ON ASTM C-890 A-16 (AASHTO HS-20-44) LOADING APPLIED IN ACCORDANCE WITH THE REQUIREMENTS OF ASTM C-890 AND ACI 318.

■ THE DESIGN IS BASED ON THE TOP SLAB
AT A DEPTH OF 0.00 FEET TO 5.00 FEET
BELOW FINISHED GRADE.

WEIGHT: 1,480 LBS
VOLUME: 9.84 CU. FT.

1. ALL LINEAR DIMENSIONS ARE $\pm 1/8"$.



DRAWN BY: BCM	DATE: 01-16-15	SCALE: NTS
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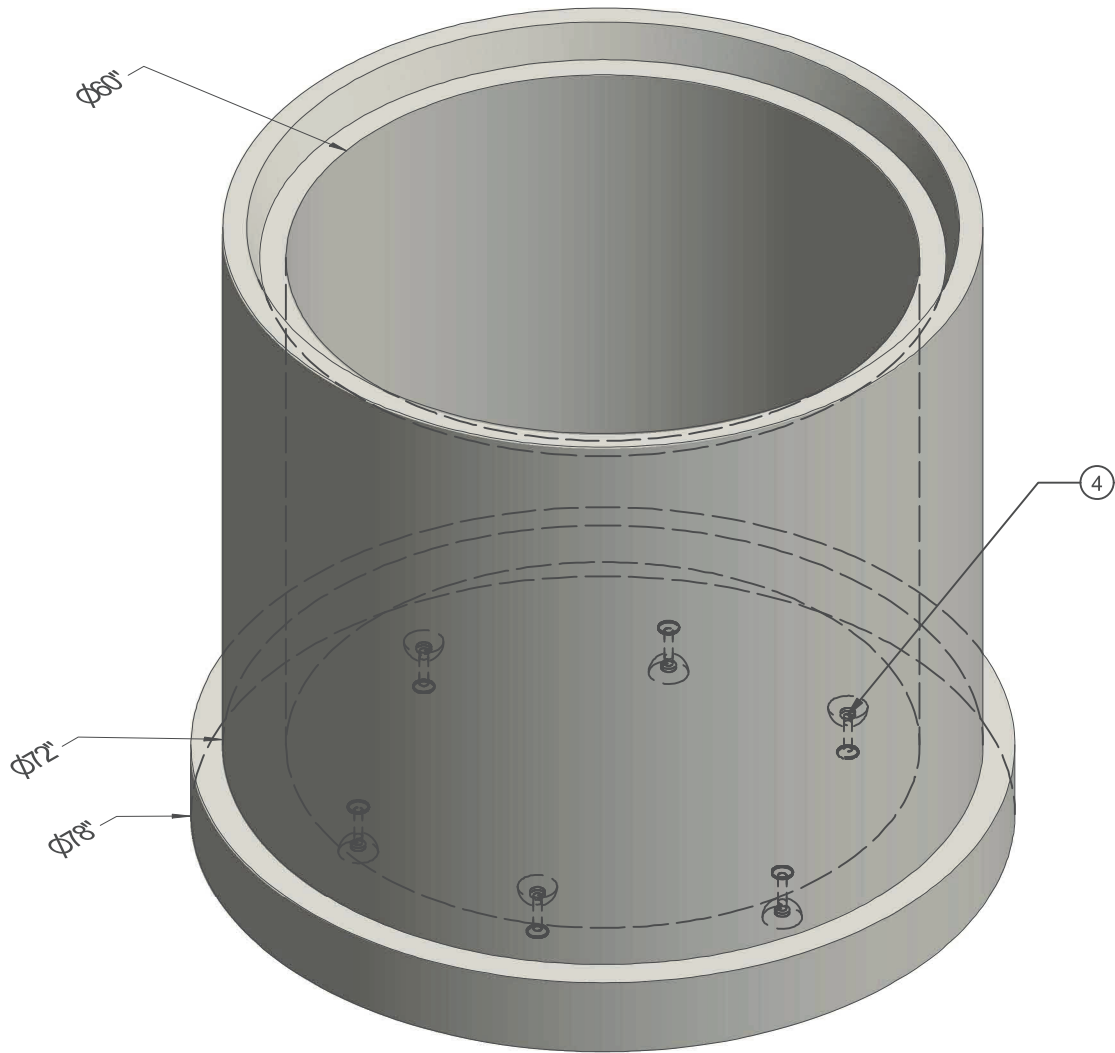
DWG. NO.: CG-FT4858-24C-8

Autocad 2015

#	DATE	DESCRIPTION	BY

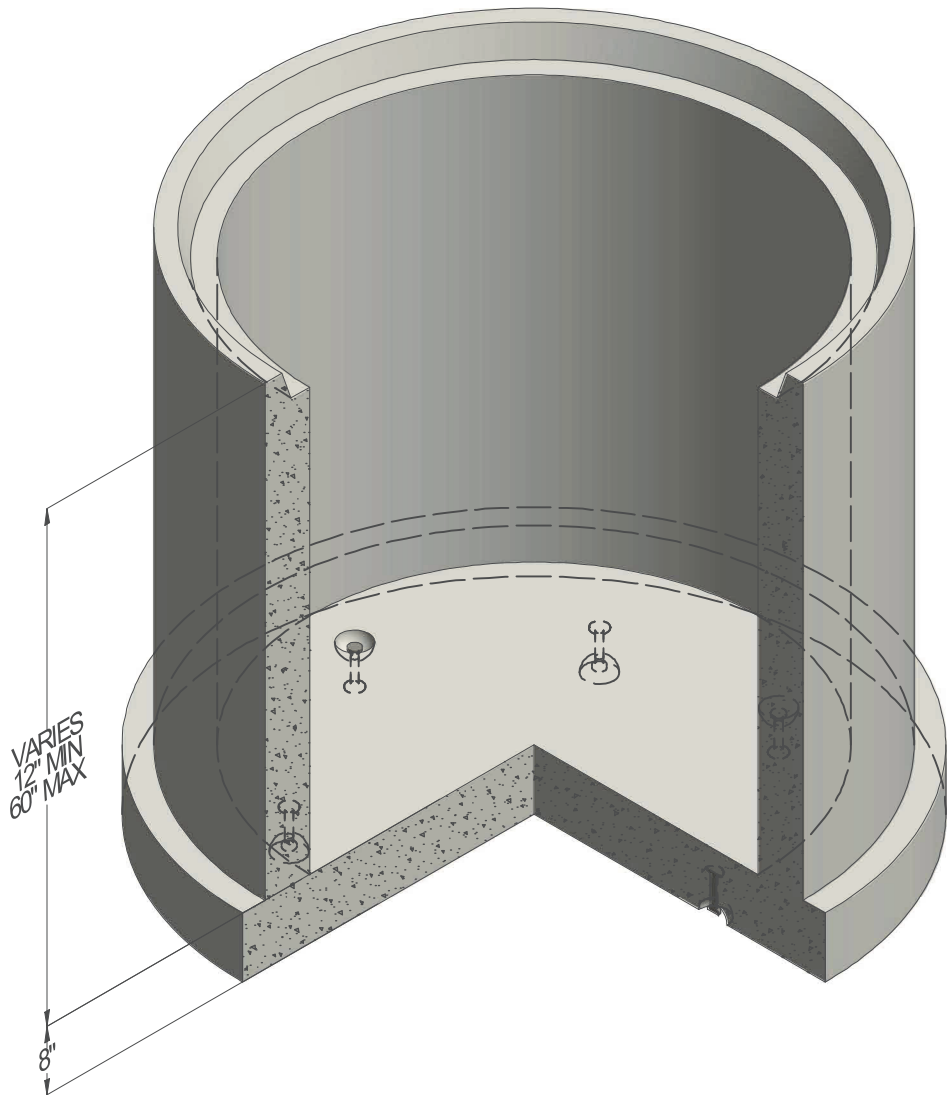
NOTES
■ FOR COMPLETE DESIGN AND PRODUCT INFORMATION CONTACT JENSEN PRECAST.

*D6072 material (for SSMH-3)



ISO REF

BILL OF MATERIALS				
ITEM	UNIT QTY	QTY	PART NUMBER	DESCRIPTION
1	64.006 ft^3	64.006 ft^3	100011428	CONCRETE MIX GENERIC 6500 PSI
2	1	1	200060381	REINFORCING MAT 200060381
3	1	1	300129551	REINFORCING MAT 300129551
4	1	6	100002874	ANCHOR DOGBONE P52 4T X 4-3/4" GALV



SECTION
REF.

LIFTING NOTES			
■ THIS CONCRETE PRODUCT IS DESIGNED TO BE LIFTED AT 2500 PSI WITH 3 LIFT CONNECTIONS EQUALLY LOADED TO 3 LIFTING POINTS AT A MINIMUM 60° SLING ANGLE, MEASURED FROM HORIZONTAL.			
BURY DEPTH:		1.0' - 30.0' to Inside of Floor	
WATER LEVEL:	0.0'	WEIGHT:	VARIES
DESIGN SPEC:	ASTM C478,ACI 318	LOADING SPEC:	A-16 (AASHTO HS20-44)
REINFORCEMENT TYPE:	Rebar,WWR Plain	LIFTING TYPE:	Dogbone

JENSEN
PRECAST



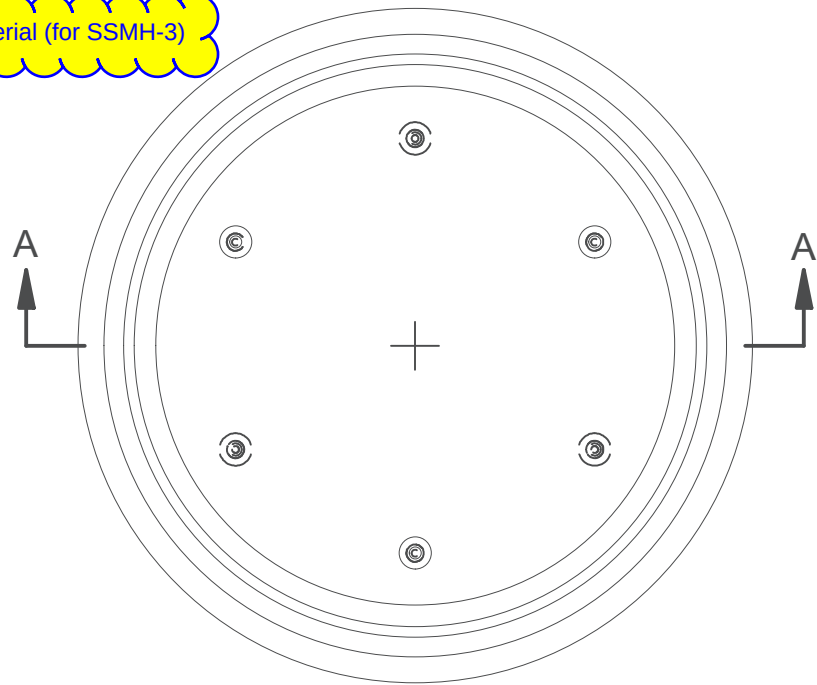
EST. 1968

WWW.JENSENPRECAST.COM

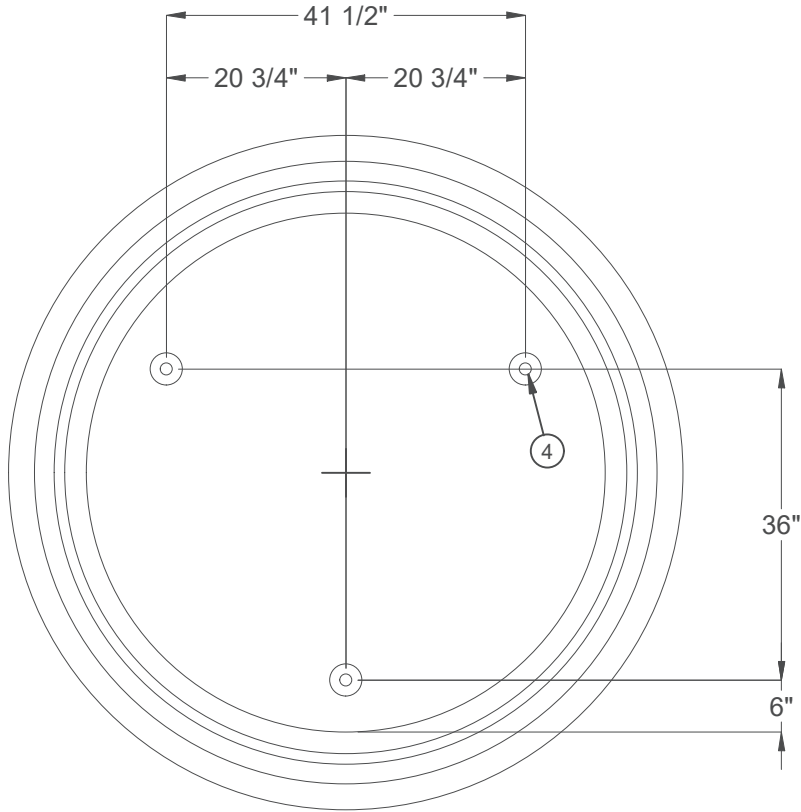
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REVISION HISTORY			
REV	DESCRIPTION	DATE	BY
1	INITIAL RELEASE	2/9/2022	BCM
DESCRIPTION:			REV:
Manhole D6072xVaries Rnd. Bottom Section GU Manually Configured			A-1
CASTING LAYOUT			
PART NUMBER:	200060382	DRAWN BY:	A.Cho
CREATED:	2/9/2022	MODIFIED:	2/11/2022
			SHEET:
			1 OF 3

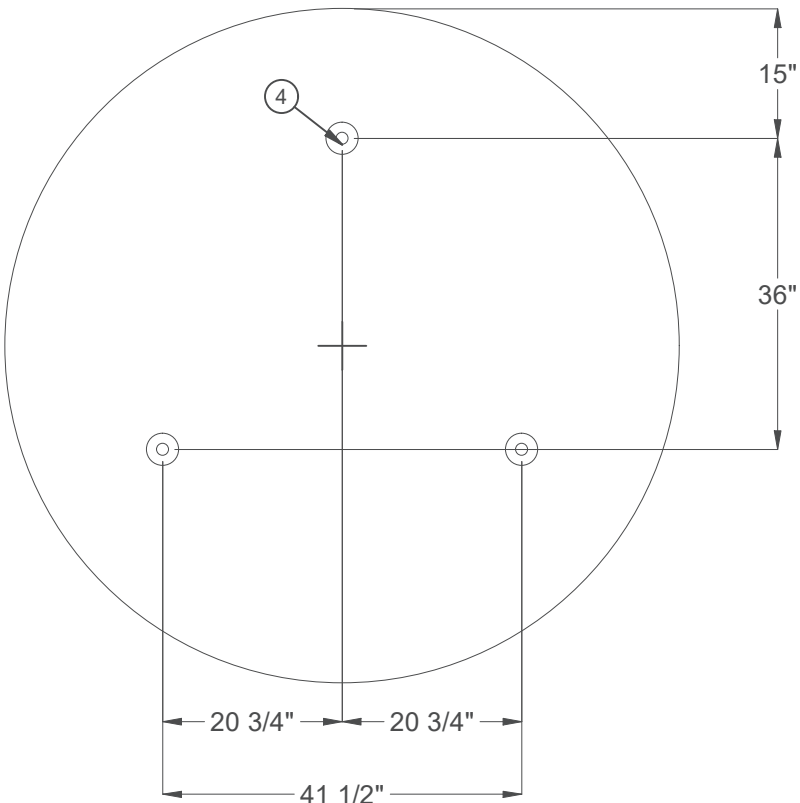
*D6072 material (for SSMH-3)



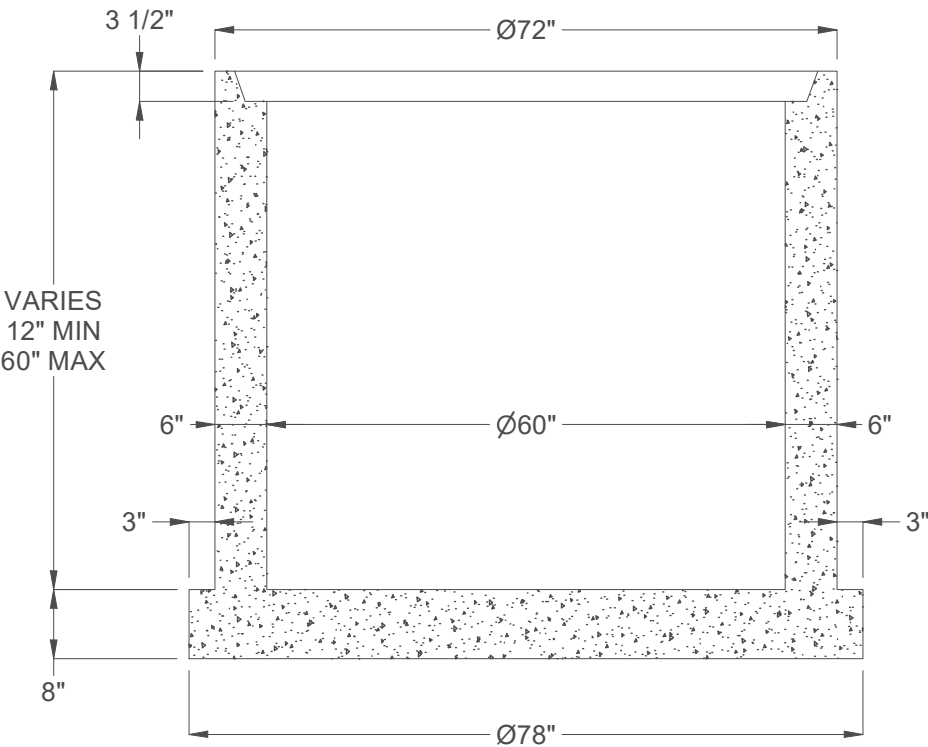
PLAN VIEW



PLAN VIEW
INSIDE FLOOR



PLAN VIEW
OUTSIDE OF FLOOR



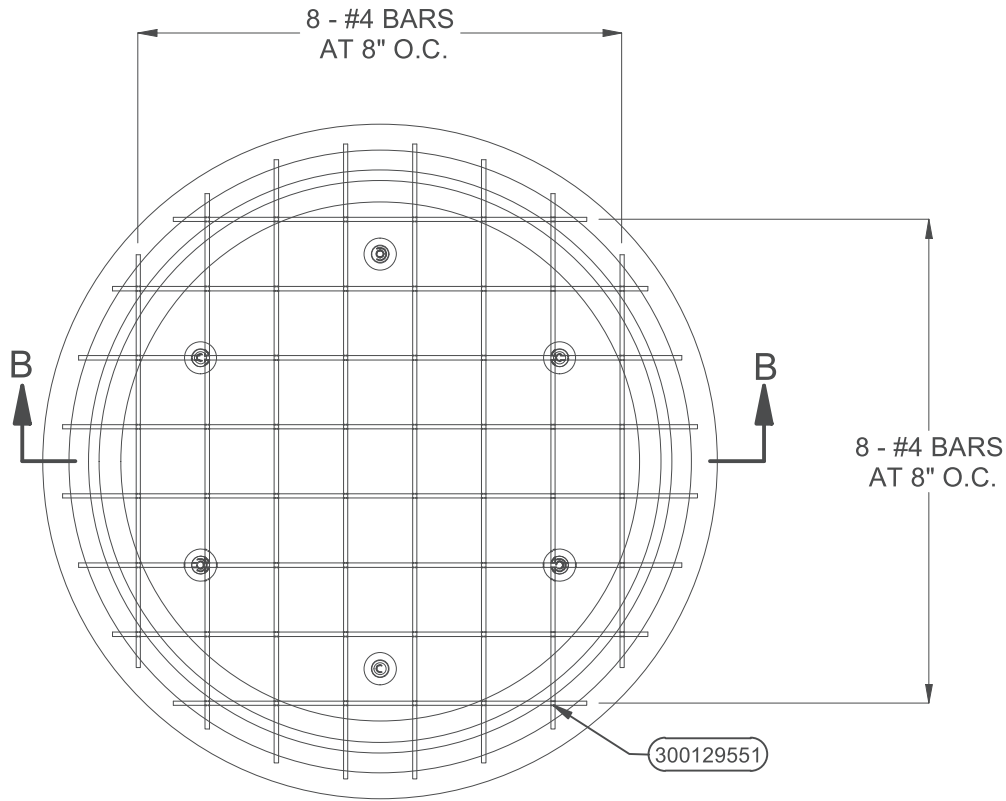
SECTION A-A

BURY DEPTH:		1.0' - 30.0' to Inside of Floor	
WATER LEVEL:		0.0'	WEIGHT: VARIES
DESIGN SPEC:		ASTM C478,ACI 318	LOADING SPEC: A-16 (AASHTO HS20-44)
REINFORCEMENT TYPE:		Rebar,WWR Plain	LIFTING TYPE: Dogbone

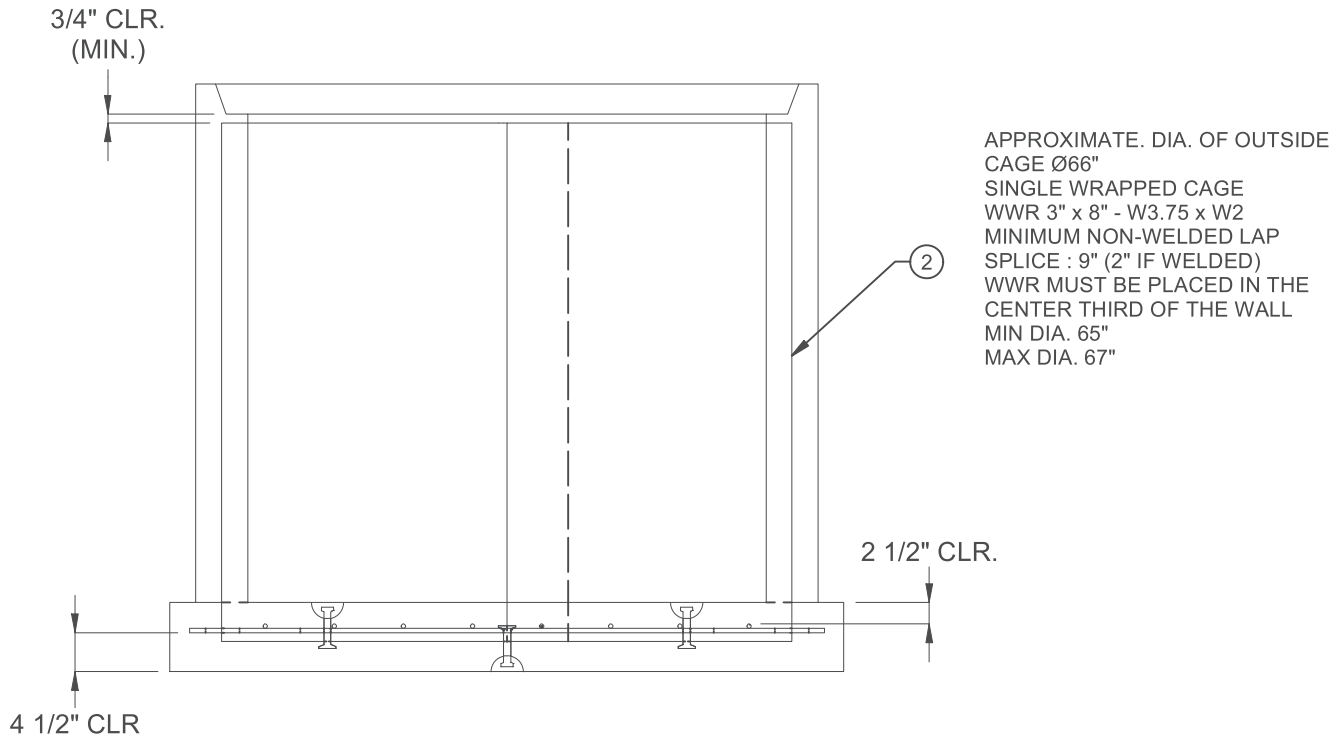
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		Manhole D6072xVaries Rnd. Bottom Section GU Manually Configured		A-1	
		CASTING DRAWING			SHEET:
		PART NUMBER:	200060382	DRAWN BY:	A.Cho
CREATED:	2/9/2022	MODIFIED:	2/14/2022	2 OF 3	

- ALL CONCRETE TO HAVE A 28 DAY COMPRESSIVE STRENGTH OF 6500 psi.
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARD.
- DEFORMED WIRE MESH SHALL COMPLY WITH ASTM A-1064 GRADE 80 STANDARDS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- BURY DEPTH 1.0' - 30.0' TO INSIDE OF FLOOR
- WATER LEVEL: 0.0'
- DESIGN SPECIFICATION: ASTM C478,ACI 318
- LOADING SPECIFICATION: A-16 (AASHTO HS20-44)
- EQUIVALENT SOIL WEIGHT DRY - 110 lbs/cu.ft.
- EQUIVALENT LATERAL PRESSURE DRY - 40 lbs/cu.ft.
- EQUIVALENT SOIL WEIGHT WET - 120 lbs/cu.ft.
- EQUIVALENT LATERAL PRESSURE WET - 80 lbs/cu.ft.

*D6072 material (for SSMH-3)



PLAN VIEW

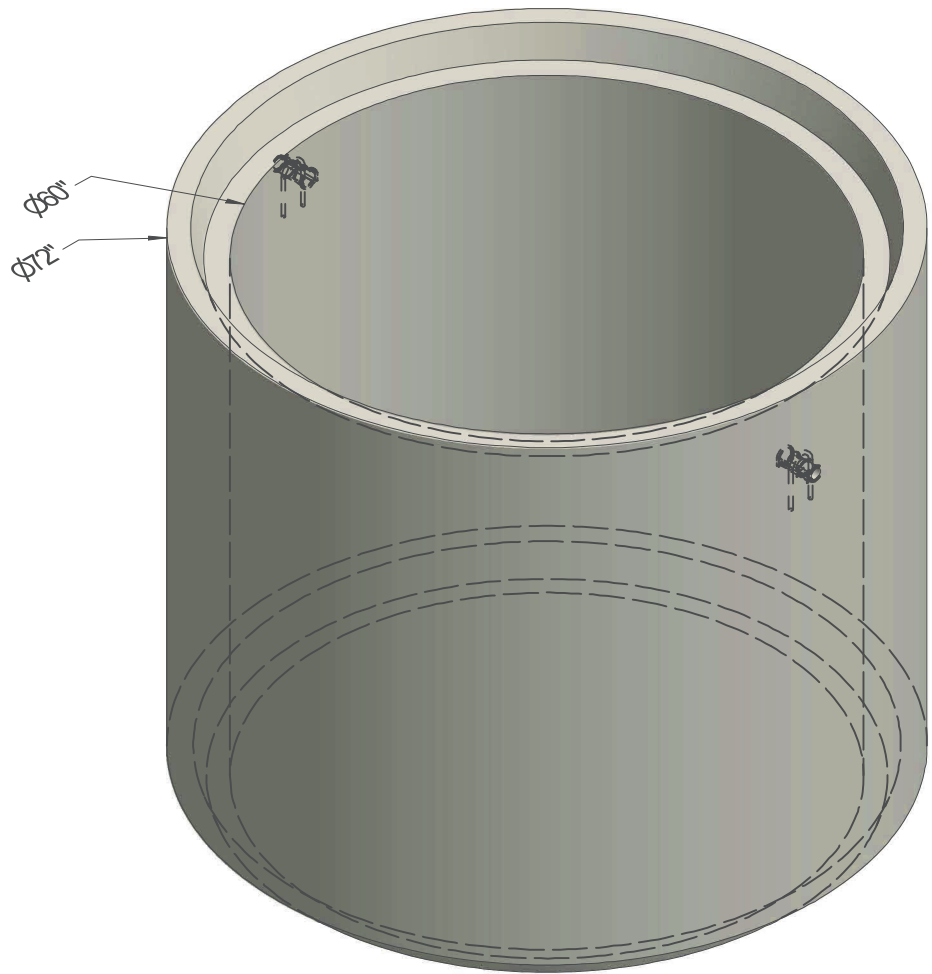


SECTION B-B

BURY DEPTH:		1.0' - 30.0' to Inside of Floor		JENSEN® PRECAST EST. 1968 WWW.JENSENPRECAST.COM	©2019 Jensen Precast- All rights reserved. All materials appearing as Jensen Precast documents and the like are proprietary work product and are protected under U.S. copyright and other laws. Unless in conjunction with business conducted with Jensen Precast, any use of Jensen Precast work product without express, written consent is prohibited, and recipient is prohibited from distributing any and all work product to non-approved third parties under penalty of civil action.	DESCRIPTION:			REV: A-1	
WATER LEVEL:		0.0'	WEIGHT:			VARIES				
DESIGN SPEC:		ASTM C478,ACI 318				LOADING SPEC:		A-16 (AASHTO HS20-44)		SHEET: 3 OF 3
REINFORCEMENT TYPE:		Rebar,WWR Plain				LIFTING TYPE:		Dogbone		
						REINFORCING CAGE REVIEW				
						PART NUMBER:		200060382	DRAWN BY:	A.Cho
						CREATED:		2/9/2022	MODIFIED:	2/11/2022

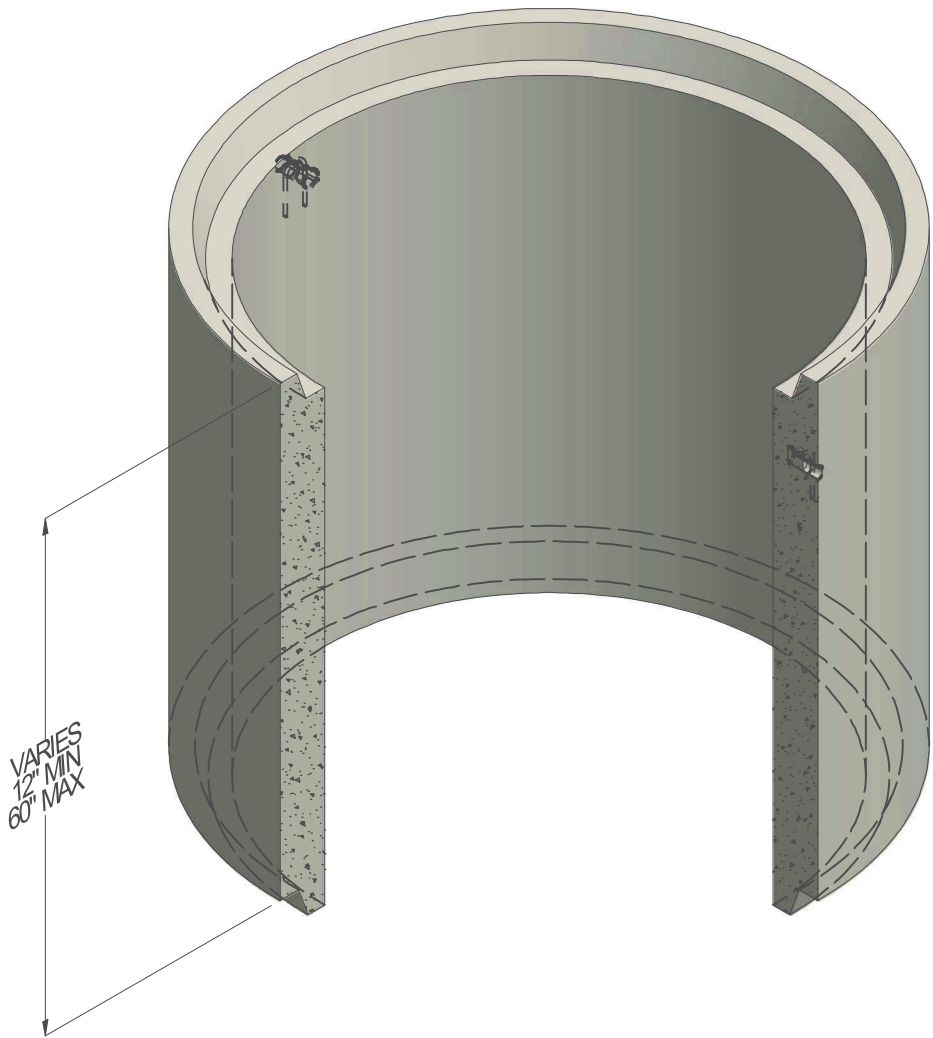
NOTES
■ FOR COMPLETE DESIGN AND PRODUCT INFORMATION CONTACT JENSEN PRECAST.

*D6072 material (for SSMH-3)



ISO REF

BILL OF MATERIALS				
ITEM	UNIT QTY	QTY	PART NUMBER	DESCRIPTION
1	43.089 ft^3	43.089 ft^3	100011428	CONCRETE MIX GENERIC 6500 PSI
2	1	2	100007288	INSERT PLASTIC LIFT PIN MH 1095A
3	13.123 in	26.247 in	100003074	REINF #3 REBAR ASTM A706/A615 GR60
4	1	1	200060410	REINFORCING MAT 200060410



SECTION
REF.

LIFTING NOTES			
■ THIS CONCRETE PRODUCT IS DESIGNED TO BE LIFTED AT 2500 PSI WITH 4 LIFT CONNECTIONS EQUALLY LOADED TO 2 LIFTING POINTS AT A MINIMUM 60° SLING ANGLE, MEASURED FROM HORIZONTAL.			
BURY DEPTH:		1.0' - 25.0' to Top	
WATER LEVEL:	0.0'	WEIGHT:	VARIES
DESIGN SPEC:	ASTM C478	LOADING SPEC:	A-16 (AASHTO HS20-44)
REINFORCEMENT TYPE:	WWR Plain	LIFTING TYPE:	Insert

JENSEN
PRECAST



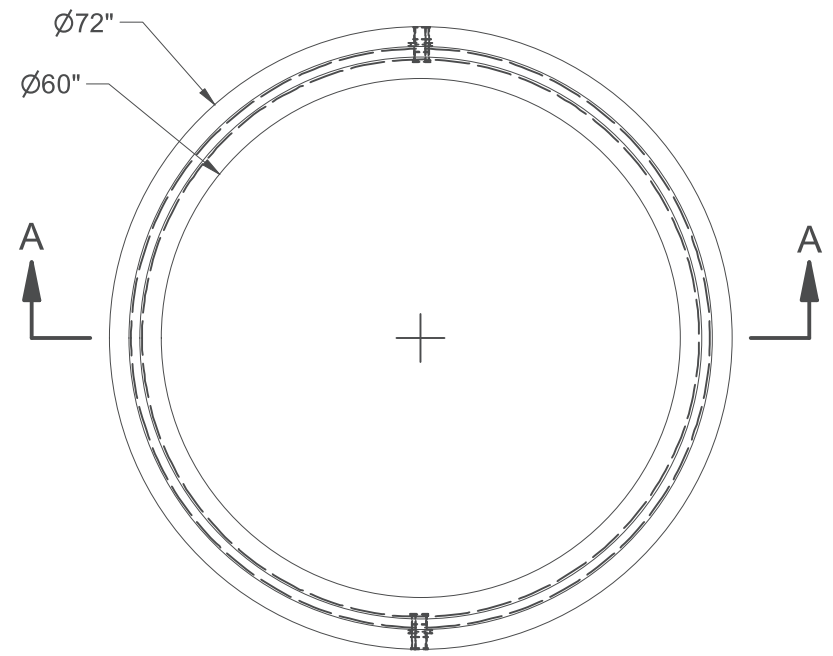
EST. 1968

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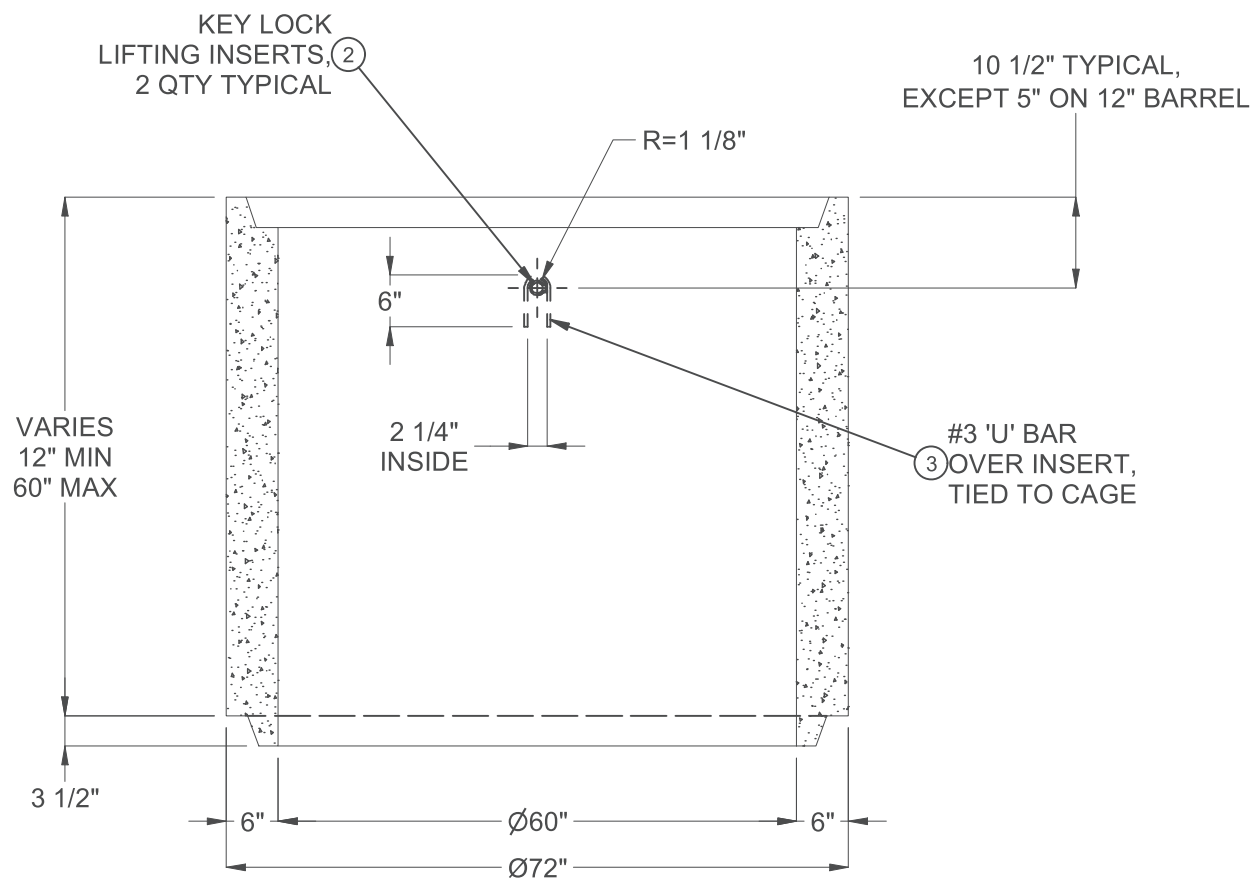
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REVISION HISTORY			
REV	DESCRIPTION	DATE	BY
1	INITIAL RELEASE	2/9/2022	BCM
DESCRIPTION:			REV:
Manhole D6072xVaries Barrel GU Manually Configured			A-1
CASTING LAYOUT			
PART NUMBER: 200060411		DRAWN BY: A.Cho	
CREATED: 2/9/2022		MODIFIED: 2/11/2022	
			SHEET:
			1 OF 3

*D6072 material (for SSMH-3)



PLAN VIEW



SECTION A-A

BURY DEPTH:		1.0' - 25.0' to Top	
WATER LEVEL:	0.0'	WEIGHT:	VARIES
DESIGN SPEC:	ASTM C478	LOADING SPEC:	A-16 (AASHTO HS20-44)
REINFORCEMENT TYPE:	WWR Plain	LIFTING TYPE:	Insert

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 EST. 1968

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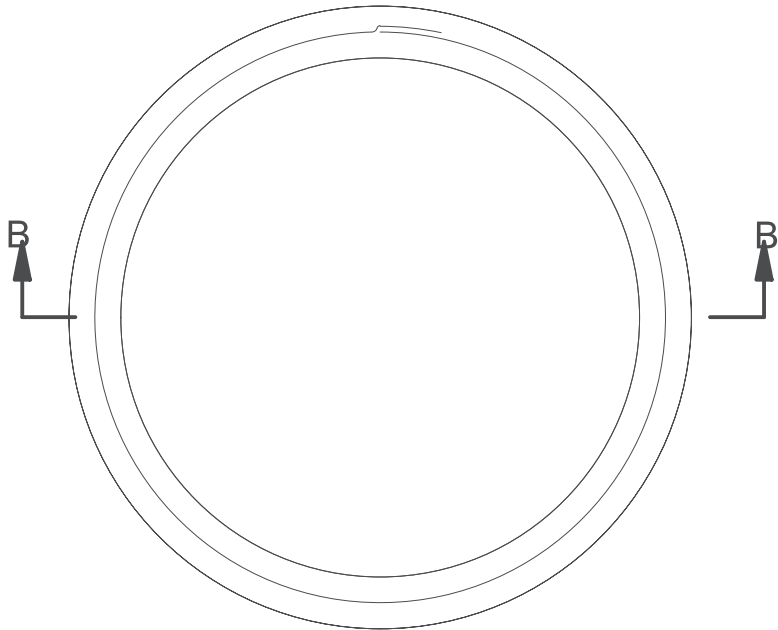
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DESCRIPTION:			
Manhole D6072xVaries Barrel GU Manually Configured			
CASTING DRAWING			
PART NUMBER:	200060411	DRAWN BY:	A.Cho
CREATED:	2/9/2022	MODIFIED:	2/11/2022

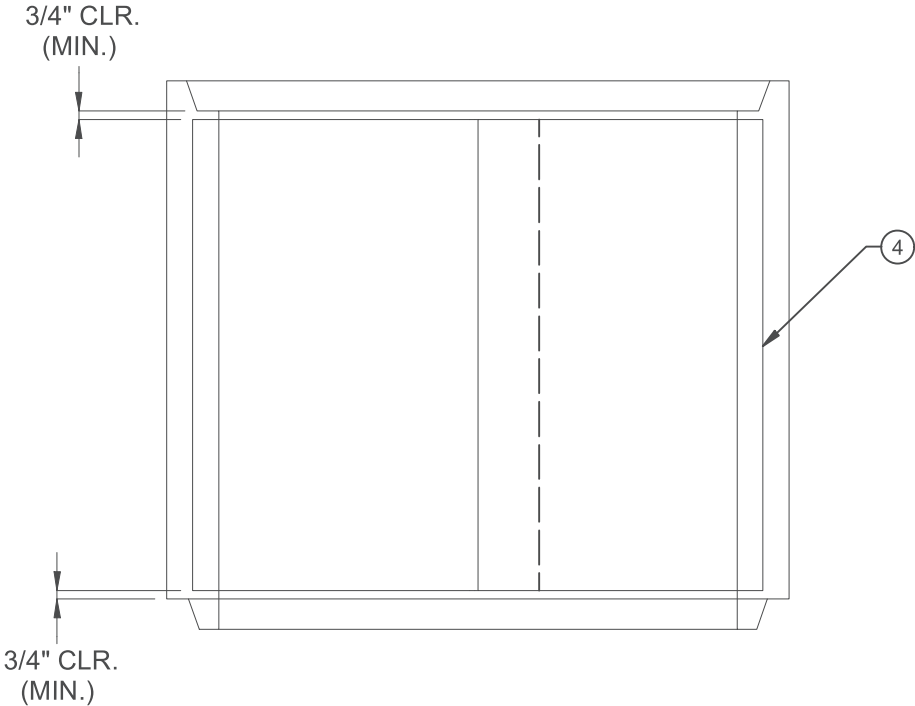
REV:	A-1
SHEET:	2 OF 3

- ALL CONCRETE TO HAVE A 28 DAY COMPRESSIVE STRENGTH OF 6500 psi.
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARD.
- DEFORMED WIRE MESH SHALL COMPLY WITH ASTM A-1064 GRADE 80 STANDARDS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- BURY DEPTH 1.0' - 25.0' TO Top
- WATER LEVEL: 0.0'
- DESIGN SPECIFICATION: ASTM C478
- LOADING SPECIFICATION: A-16 (AASHTO HS20-44)
- EQUIVALENT LATERAL PRESSURE DRY - 40 lbs/cu.ft.
- EQUIVALENT LATERAL PRESSURE WET - 80 lbs/cu.ft.

*D6072 material (for SSMH-3)



PLAN VIEW
(JOINT HIDDEN FOR CLARITY)

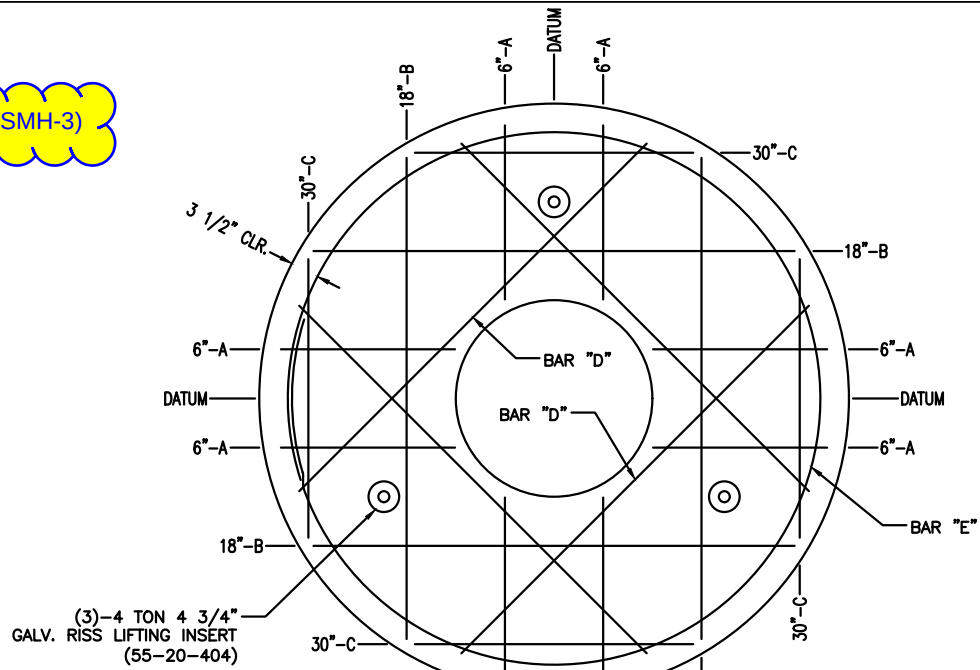


APPROXIMATE. DIA. OF OUTSIDE
CAGE Ø66"
SINGLE WRAPPED CAGE
WWR 3" x 8" - W3.75 x W2
MINIMUM NON-WELDED LAP
SPLICE : 9" (2" IF WELDED)
WWR MUST BE PLACED IN THE
CENTER THIRD OF THE WALL
MIN DIA. 65"
MAX DIA. 67"

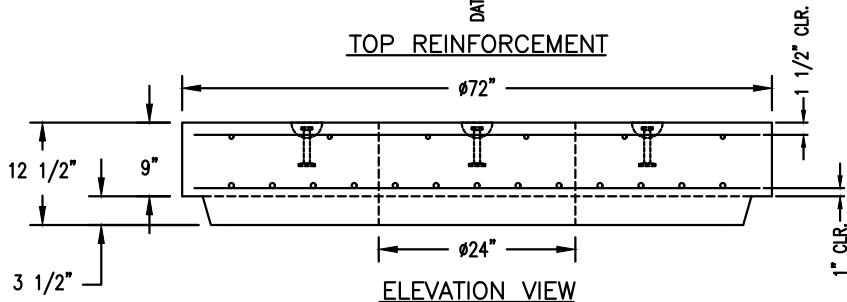
SECTION B-B

BURY DEPTH:		1.0' - 25.0' to Top		JENSEN® PRECAST EST. 1968 WWW.JENSENPRECAST.COM	DESCRIPTION:		Manhole D6072xVaries Barrel GU Manually Configured		REV:
WATER LEVEL:	0.0'	WEIGHT:	VARIES						A-1
DESIGN SPEC:	ASTM C478	LOADING SPEC:	A-16 (AASHTO HS20-44)		REINFORCING CAGE REVIEW				SHEET:
REINFORCEMENT TYPE:	WWR Plain	LIFTING TYPE:	Insert		PART NUMBER:	200060411	DRAWN BY:	A.Cho	3 OF 3
				CREATED:		2/9/2022	MODIFIED:	2/11/2022	

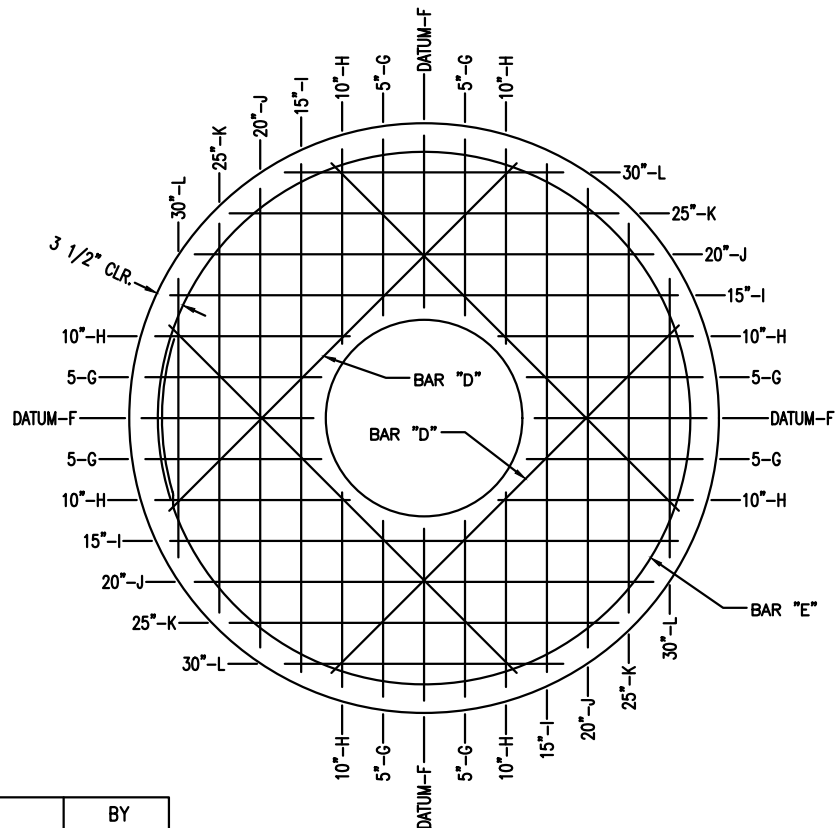
*D6072 material (for SSMH-3)



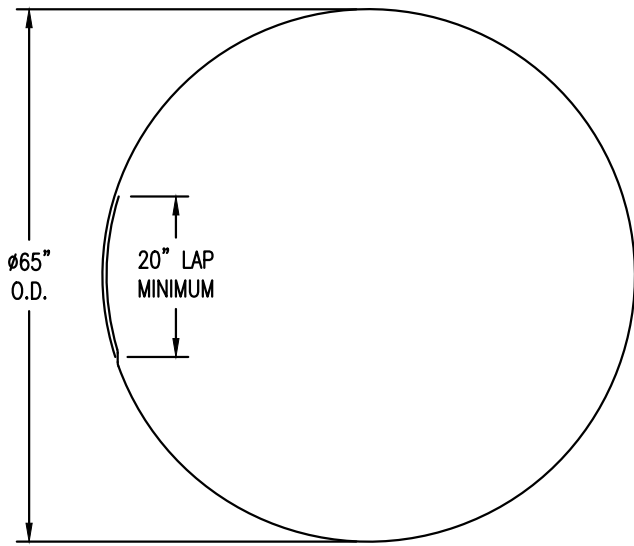
TOP REINFORCEMENT



ELEVATION VIEW



BOTTOM REINFORCEMENT



BEND DETAIL "E"

BILL OF MATERIALS

Item	Qty.	Description	Ft.	Wt.	L1	CENTER	L2
A	8	# 4 REBAR x 21 1/4	14.2	9.5		21 1/4	
B	4	# 4 REBAR x 58 3/4	19.6	13.1		58 3/4	
C	4	# 4 REBAR x 34	11.3	7.6		34	
D	8	# 4 REBAR x 60	40.0	26.7		60	
E	2	# 3 REBAR x 224 1/2	37.4	14.1	SEE BEND DETAIL "E"		
F	4	# 5 REBAR x 21	7.0	7.3		21	
G	8	# 5 REBAR x 21 1/2	14.3	14.9		21 1/2	
H	8	# 5 REBAR x 23 3/4	15.8	16.5		23 3/4	
I	4	# 5 REBAR x 62	20.7	21.6		62	
J	4	# 5 REBAR x 56	18.7	19.5		56	
K	4	# 5 REBAR x 47 1/2	15.8	16.5		47 1/2	
L	4	# 5 REBAR x 34	11.3	11.8		34	
TOTALS			226.2	179.0			

3 37 4 85 5 104 6 0

NOTES:

- ALL CONCRETE TO HAVE A 28 DAY COMPRESSIVE STRENGTH OF 4500 psi.
- REINFORCING STEEL SHALL COMPLY WITH ASTM A615 GRADE 60 OR ASTM A706 GRADE 60. BAR BENDING AND PLACEMENT SHALL COMPLY WITH THE LATEST ACI STANDARDS.
- EQUIVALENT AREA $F_y = 70000$ psi ASTM A-1064 REINFORCING CAGES MAY BE UTILIZED IN LIEU OF GRADE 60 REINFORCING BARS.
- LIFTING INSERTS FOR HANDLING SHALL BE INSTALLED PER MANUFACTURER'S REQUIREMENTS.
- STRUCTURAL DESIGN IS BASED ON ASTM C-890 A-16 (H20) LOADING
- WATER TABLE ASSUMED AT FINISHED GRADE
SOIL UNIT WEIGHT - 120 lbs/cu.ft.
EQUIVALENT LATERAL PRESSURE DRY - 40 lbs/cu.ft.
EQUIVALENT LATERAL PRESSURE WET - 80 lbs/cu.ft.
- THE DESIGN IS BASED ON TOP AT 0.0' TO 5.0' BELOW FINISHED GRADE.
- CONCRETE WEIGHT: 3,750 LBS

CALCULATION REFERENCE: D6072-TS09-5-CALC

JENSEN
PRECAST

91-110 HANUA ST SUITE # 202, KAPOLEI, HAWAII 96707

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DRAWN BY:
BCM

DATE:
6-13-2014

SCALE:
NTS

REBAR FOR 60" DIA. FLAT TOP W/
24" DIA. CONCENTRIC OPENING

DWG. NO.:

CG-FT6024C

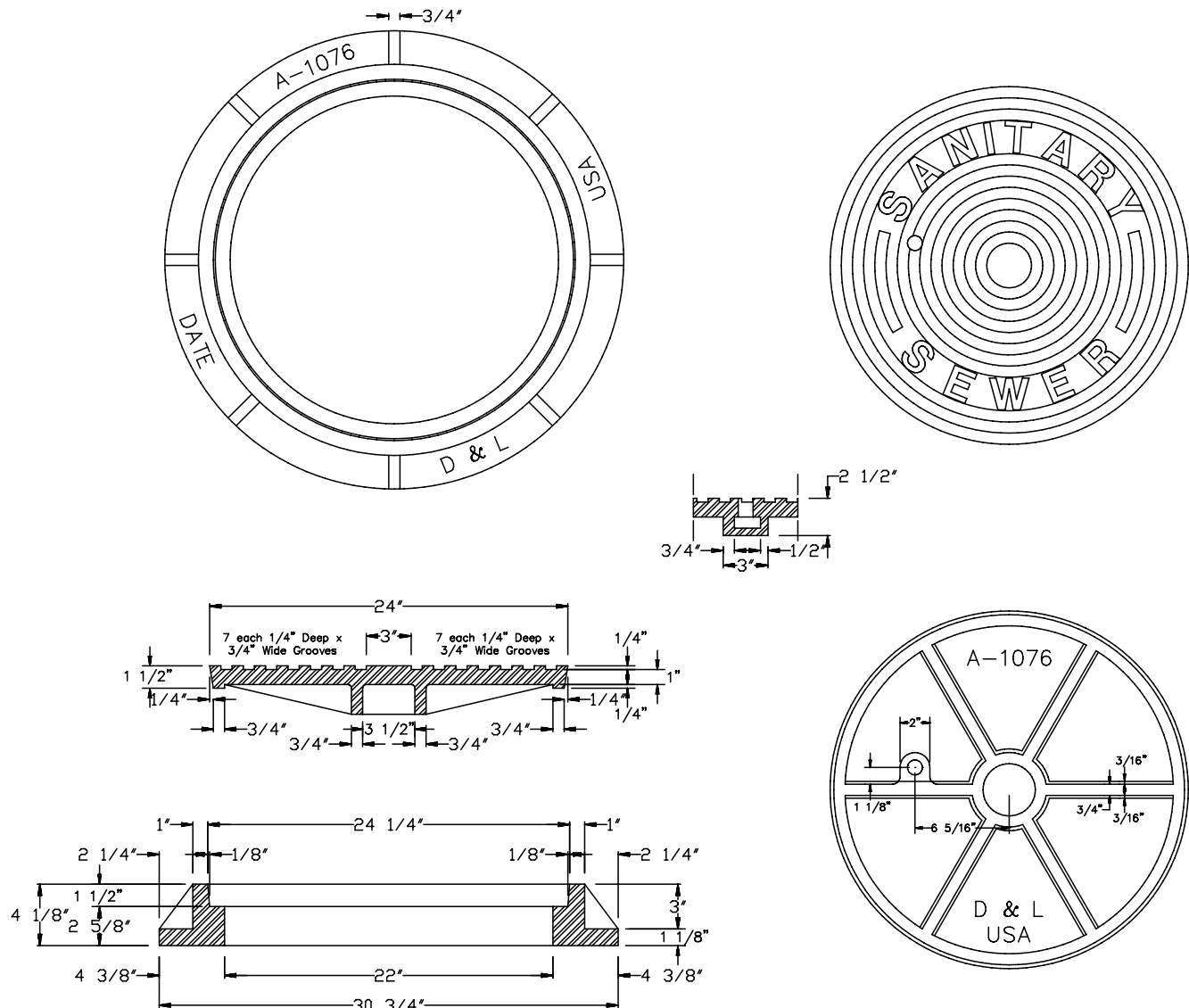
Part 100008494

AutoCAD 2014

#	DATE	DESCRIPTION	BY

A-1076

Manhole Ring and Cover



Note: Different municipality names available on cover

CAST IRON to conform to ASTM
A-48, CLASS 35B
H-20 Wheel Loading

D&L No. A-1076

Est. weight of Cover 150 lbs.
Est. weight of Ring 205 lbs.



D&L Foundry
P.O. Box 1319
Moses Lake, WA 98837

Phone:
(509)
765-7952

Fax:
(509)
765-8124

Designation
A-1076

Date of Drawing
OCT 1994

Prepared by
D&L Supply

Scale
1" = 11 1/2"

Sheet Number
1

Total Sheets
1

ConSeal™ CS-102

Butyl Rubber Sealant



Butyl Rubber Sealant for All Precast Concrete Structures - Meets ASTM C-990

Applications

For concrete joints in: Manholes, Concrete Pipe, Vaults, Box Culverts, Septic Tanks, and Vertical Panel Structures. **Not intended for use in expansion joints or joints that move.**

Sealing Properties

- Provides permanently flexible watertight joints.
- Low to high temperature workability: 30°F to 120°F (-1°C to +48°C)
- Rugged service temperature: -30°F to +200°F (-34°C to +93°C)
- Excellent chemical and mechanical adhesion to clean dry surfaces.
- Greater cohesive and adhesive strengths.
- Sealed joints will not shrink, harden or oxidize upon aging.
- Controlled flow resistance for application ease.
- No priming normally necessary. When confronted with difficult installation conditions, such as wet concrete or temperatures below 40°F (4°C), priming the concrete will improve the bonding action. Consult Concrete Sealants for the proper primer to meet your application.

Hydrostatic Strength

ConSeal CS-102 meets the hydrostatic performance requirement as set forth in ASTM C-990 section 10.1 (Performance requirement: 10psi for 10 minutes in straight alignment – in plant, quality control test for joint materials.)

Specifications

ConSeal CS-102 meets or exceeds all of the requirements of Federal Specification SS-S-210 (210-A), AASHTO M-198B, and ASTM C-990-91.

Physical Properties

Description

	Spec	Required	CS-102
Color			Black
Specific Gravity, 77°F	ASTM D71	1.15-1.50	1.25
Ductility, 77°F	ASTM D113	5.0 min.	10
Penetration, cone 77°F (25°C), 150 gm, 5 sec.	ASTM D217	50-100 mm	55-60 mm
Penetration, cone 32°F (0°C), 150 gm, 5 sec.	ASTM D217	40 mm min.	40-65 mm
Flash Point, C.O.C., °F	ASTM D92	350°F min.	450°F
Fire Point, C.O.C., °F	ASTM D92	375°F min.	475°F

Don't Just Seal It, ConSeal It!

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Concrete Sealants, Inc. 9325 State Route 201 ■ Tipp City, OH 45371 ■ P.O. Box 176 ■ New Carlisle, OH 45344
P. 937.845.8776 F. 937.845.3587 Toll Free 800.332.7325 ■ www.conseal.com



ConSeal™ CS-102

Butyl Rubber Sealant



Butyl Rubber Sealant for All Precast Concrete Structures - Meets ASTM C-990

Chemical Composition

Description

	Spec	Required	CS-102
Hydrocarbon plastic content % by weight	ASTM D4 (mod.)	50% min.	51%
Inert mineral filler % by weight	AASHTO T111	30% min.	35%
Volatile Mater % by weight	ASTM D6	2% max.	1.2%
Recycled Content, % by weight			
Post Consumer:			8.41%
Post Industrial:			10.85%

Immersion Testing

30-Day Immersion Testing: No visible deterioration when tested in 5% Caustic Potash, 5% Hydrochloric Acid, 5% Sulfuric Acid, and 5% saturated Hydrogen Sulfide.

One Year Immersion Testing: No visible deterioration when tested in 5% Formaldehyde, 5% Formic Acid, 5% Sulfuric Acid, 5% Hydrochloric Acid, 5% Sodium Hydroxide, 5% Hydrogen Sulfide, and 5% Potassium Hydroxide.

Limited Warranty

This information is presented in good faith, but we cannot anticipate all conditions under which this information and our products, or the products of other manufactures in combination with our products, may be used. We accept no responsibility for results obtained by the application of this information or the safety and suitability of our products, either alone or in combination with other products. Users are advised to make their own tests to determine the safety and suitability of each such product or product combinations for their own purposes. It is the **users' responsibility** to satisfy himself as to the suitability and completeness of such information for this own particular use. We sell this product without warranty, and buyers and users assume all responsibility and liability for loss or damage arising from the handling and use of this product, whether used alone or in combination with other products.

Don't Just Seal It, ConSeal It!

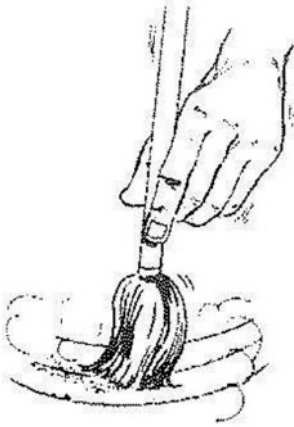
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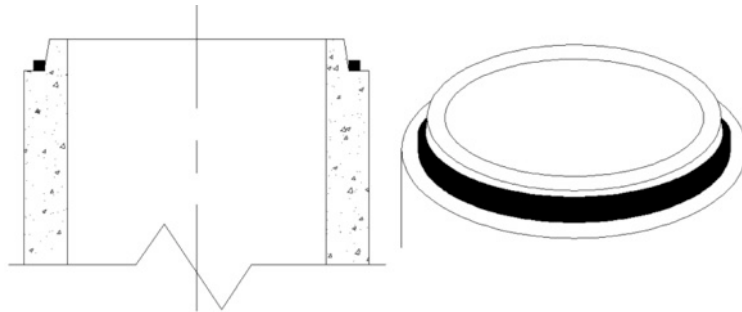


Installation Instructions

Manhole



Clean the concrete joint surfaces by brushing off all loose concrete, dust, and dirt.



Apply ConSeal all around the outside of the spigot. Press into place.

The following procedures should be followed for optimum sealant performance.

- Clean the bell and spigot joint surfaces with a stiff bristle brush.
- Remove any dirt, debris, flashing, or concrete high points, which could keep the joint from coming together.
- If necessary, a joint primer (i.e. CS-50 or CS-75) can be applied to improve sealant adhesion.
- When using a primer, allow the primer to dry before placing sealant. The primer will normally be dry within 15-20 minutes. CS-50 will be dull black, but not tacky. CS-75 will be orange and tacky.
Note: CS-50 primer can be applied at the plant several days in advance of the set. CS-75 must be applied the day of the installation.
- Place sealant on the tip of the spigot in the correct location (see image above).
- Join the sealant into one continuous strand by kneading the ends together where they meet.
DO NOT STRETCH THE SEALANT. DO NOT OVERLAP THE SEALANT.
- Once the sealant is installed, do not:
 - Step on the sealant
 - Allow dirt or debris to fall on the sealant
- A **minimum** compression of 50% is required. Greater than 50% compression is optimal. This will vary according to the ambient temperature and the weight being applied.

Disclaimer: This publication is to assist users to understand the proper use of ConSeal's products. *Contact ConSeal's technical staff for installation instructions that meet your specific requirement.* Concrete Sealants, Inc. does not warranty any improper use of its products.

Don't Just Seal It, ConSeal It!

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